Semiconductor Devices

Physics and Technology

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3rd Edition

CHAPTER 2: Carrier Transport Phenomena

2.1 CARRIER DRIFT

2.1.1 Mobility

* Kinetic energy of the electrons:

→ $\frac{1}{2} m_n v_{th}^2 = \frac{3}{2} kT$

m_n: Effective mass of electrons

v_{th}: Average thermal velocity of electrons

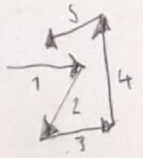
Theorem for equipartition of energy: $\frac{1}{2} kT$ units per degree of freedom

→ $v_{th} = \sqrt{\frac{3kT}{m_n}}$

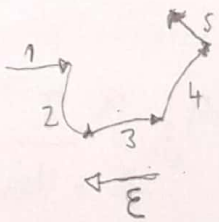
At 300K: For Si and gallium Arsenide:

$v_{th} \approx 10^7 cm/s$

* Random thermal motion



Applied electric field:



For the mean free path of: $10^{-5} cm = 0,1 \mu m$:

Mean free time $\tau_c \approx \frac{10^{-5} cm}{v_{th}} = \frac{10^{-5} cm}{10^7 cm/s} = 10^{-12} seg$

$= 1 ps$

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* Momentum applied to an electron = Momentum gained

→ (Force) x (time) = $m_n v_n$

$-q\mathcal{E}\tau_c = m_n v_n$

v_n : Electron drift velocity

$$v_n = -\left(\frac{q\tau_c}{m_n}\right)\mathcal{E}$$

Steady state: All momentum gained between collisions is lost to the lattice in the collision

Electron mobility:

→ $\mu_n = \frac{q\tau_c}{m_n}$

Thus:

$$\boxed{v_n = -\mu_n \mathcal{E}}$$

For holes in the valance band:

$$\boxed{v_p = \mu_p \mathcal{E}}$$

→ $\mu_p = \frac{q\tau_s}{m_p}$

v_p : Hole drift velocity

→ $[\mu] = \frac{[v]}{[\mathcal{E}]} = \frac{cm/s}{V/cm} = \frac{cm^2}{V\,s}$

Mobility due to lattice scattering:

$\mu_L \propto T^{-3/2}$ ← Lattice vibrations $\propto T$

$\mu_L \propto \frac{1}{T^{3/2}}$

Mobility due to impurity scattering:

$\mu_I \propto \frac{T^{3/2}}{N_T}$

N_T : Total impurity concentration

The number of collisions taking place in a unit time:

$$\frac{1}{\tau_c} = \frac{1}{\tau_{c,\,lattice}} + \frac{1}{\tau_{c,\,impurity}}$$

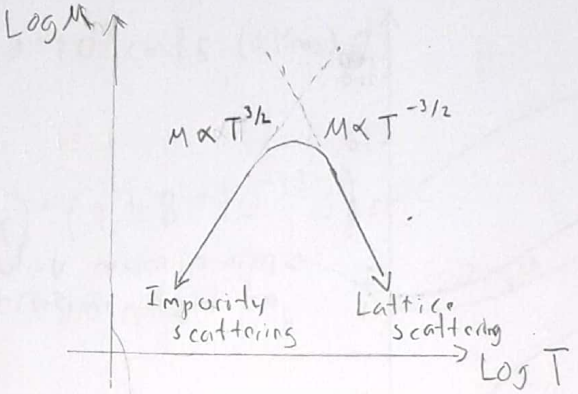
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So!

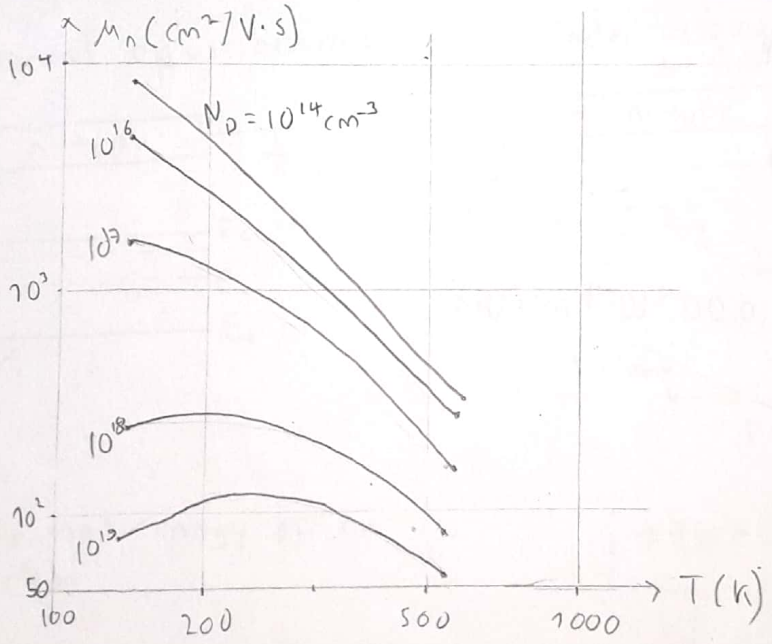
$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I} \qquad \rightarrow \mu = \frac{q\tau}{m}$$

$$\frac{1}{\mu} = \frac{m}{q\tau}$$

Theoretical temperature dependence of μ:



μ in silicon for various donor concentrations:



For lightly doped samples ($\sim 10^{14}\,cm^{-3}$):

$\rightarrow \mu_L \propto T^{-3/2}$ $\qquad$ $\rightarrow \mu_I \propto T^{3/2}/N_T$

$$\frac{1}{\mu} = \frac{1}{\mu_L} + \frac{1}{\mu_I} \qquad \frac{1}{\mu_L} \propto T^{3/2} \qquad \frac{1}{\mu_I} \propto \frac{N_D}{T^{3/2}}$$

↓

Lattice scattering dominates

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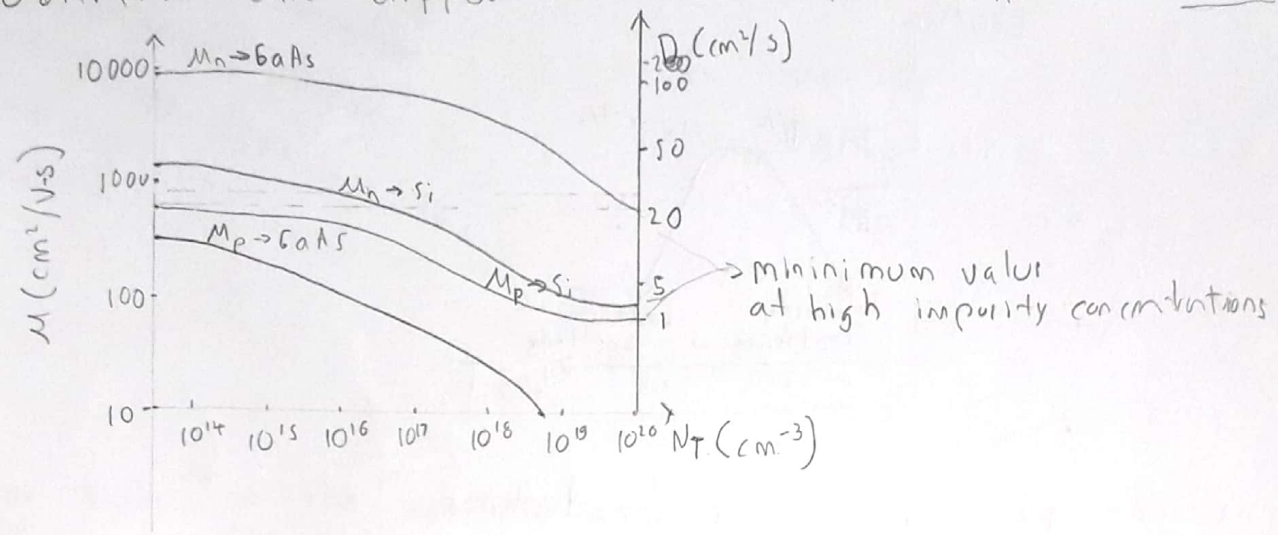
For heavily doped samples ($\sim 10^{19}\,cm^{-3}$)

$$\frac{1}{\mu} = \frac{1}{\mu_L}^{0} + \frac{1}{\mu_I} \quad \rightarrow \frac{1}{\mu_I} \propto \frac{N_D}{T^{3/2}}$$

↓

To Low temperature

Mobilities and diffusitives in Si and GaAs at 300K:



$m_n < m_p$

↓

$\mu_n > \mu_p$

EXAMPLE 1

$\mu_n = 1000\ cm^2/V\text{-}s = 1000 \times 10^{-4}\ m^2/V\text{-}s$

$\tau_c = ?$

Mean Free path $= \lambda = ?$

$m_n = 0,26\, m_0$

$$\rightarrow \mu_n = \frac{q\,\tau_c}{m_n}$$

$$\tau_c = \frac{\mu_n m_n}{q} = \frac{(0,26 \times 9,1 \times 10^{-31}\ kg)(1000 \times 10^{-4}\ m^2/V\text{-}s)}{1,6 \times 10^{-19}\ C}$$

$$\tau_c = 1,48 \times 10^{-13}\ s$$

$$\rightarrow \frac{kg\,m^2}{V\text{-}s\,C} = \frac{kg\,m^2}{\frac{J}{C}\text{-}s\,C}$$

$$= \frac{kg\,m^2}{\frac{kg\,m^2}{s^2}\,s} = s\,cg$$

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Thermal velocity:

$$\frac{1}{2} m_n v_{th}^2 = \frac{3}{2} kT$$

$$v_{th} = \sqrt{\frac{3kT}{m_n}} = \sqrt{\frac{3(1,38 * 10^{-23} J/K)(300K)}{(0,26 * 9,1 * 10^{-31} kg)}}$$

$$v_{th} = 2,29 * 10^5 m/s$$

$$= 2,29 * 10^7 cm/s$$

→ $l = v_{th} \cdot T_c$

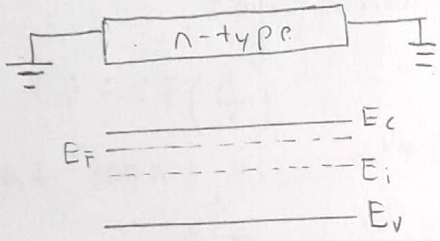
$$l = (2,29 * 10^7 cm/s) * (1,48 * 10^{-13} s)$$

$$\lambda = 3,37 * 10^{-6} cm = 33,7 nm$$

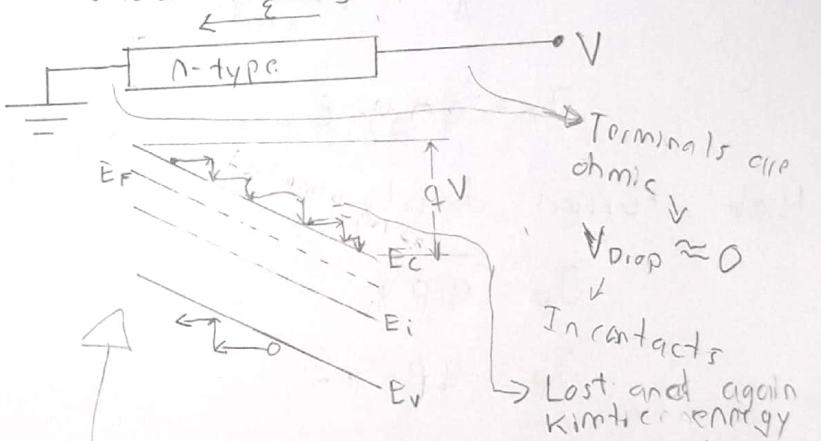
2.1.2 Resistivity:

Conduction process in an n-type semiconductor:

at thermal equilibrium:



Under a biasing condition:



→ E_c = Potential energy of an electron

→ Force from field = − gradient of electron potential energy

→ E_F, E_i or E_v are parallel to E_c:

$$\varepsilon = \frac{1}{q}\frac{dE_c}{dx} = \frac{1}{q}\frac{dE_F}{dx} = \frac{1}{q}\frac{dE_v}{dx}$$

$$\varepsilon = \frac{1}{q}\frac{dE_i}{dx}$$

$$(+q\varepsilon) = -\left(\frac{dE_c}{dx}\right)$$

↘ Negative → ε negative → ε

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ψ: Electrostatic potential:

$$\rightarrow \mathcal{E} = -\frac{d\psi}{dx}$$

$$\left(\frac{1}{q}\frac{dE_i}{dx}\right) = -\frac{d\psi}{dx}$$

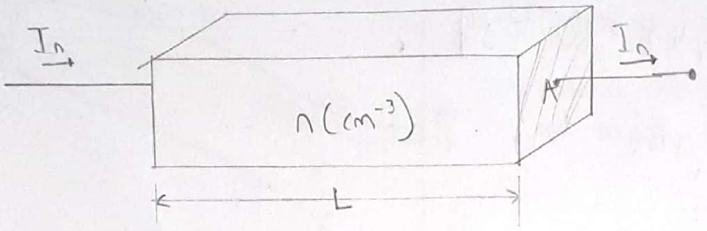
$$\psi = -\frac{E_i}{q}$$

($\mathcal{E}$ constant)

$$\rightarrow \mathcal{E} = \frac{V}{L}$$

(L: Sample length)

* Current conduction in a uniformly doped semiconductor:



Electron current density:

$$J_n = \frac{I_n}{A} = \sum_i \frac{(-qv_i)}{cm^3} = -qnv_n = -qn(-\mu_n \mathcal{E})$$

$$J_n = qn\mu_n \mathcal{E}$$

Hole current density:

$$J_p = qpv_p$$

$$J_p = qp\mu_p \mathcal{E}$$

Total current density:

$$J = J_n + J_p$$

$$J = q(n\mu_n + p\mu_p)\mathcal{E}$$

$$J = \sigma \mathcal{E}$$

$\rightarrow \sigma$ = Conductivity $= q(n\mu_n + p\mu_p)$

Resistivity of the semiconductor:

$$\rightarrow \boxed{\rho = \frac{1}{\sigma} = \frac{1}{q(n\mu_n + p\mu_p)}}$$

For an n-type semiconductor (since $n \gg p$):

$$\rho = \frac{1}{qn\mu_n}$$

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For an p type semiconductor: (since $p >> n$):

$$\rho = \frac{1}{q p \mu_p}$$

* For a thin semiconductor sample:

Thickness << diameter

$w << d$

$$\rightarrow \rho = \frac{V}{I} \cdot W \cdot CF$$

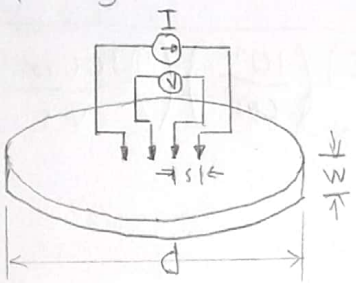
$$\rightarrow [\rho] = \frac{1}{C \frac{1}{cm^3} \left(\frac{cm^2}{Vs}\right)}$$

$$[\rho] = \frac{Vs\ cm}{C} = \frac{V}{A} cm$$

$$[\rho] = \Omega\ cm$$

→ CF: Correction Factor

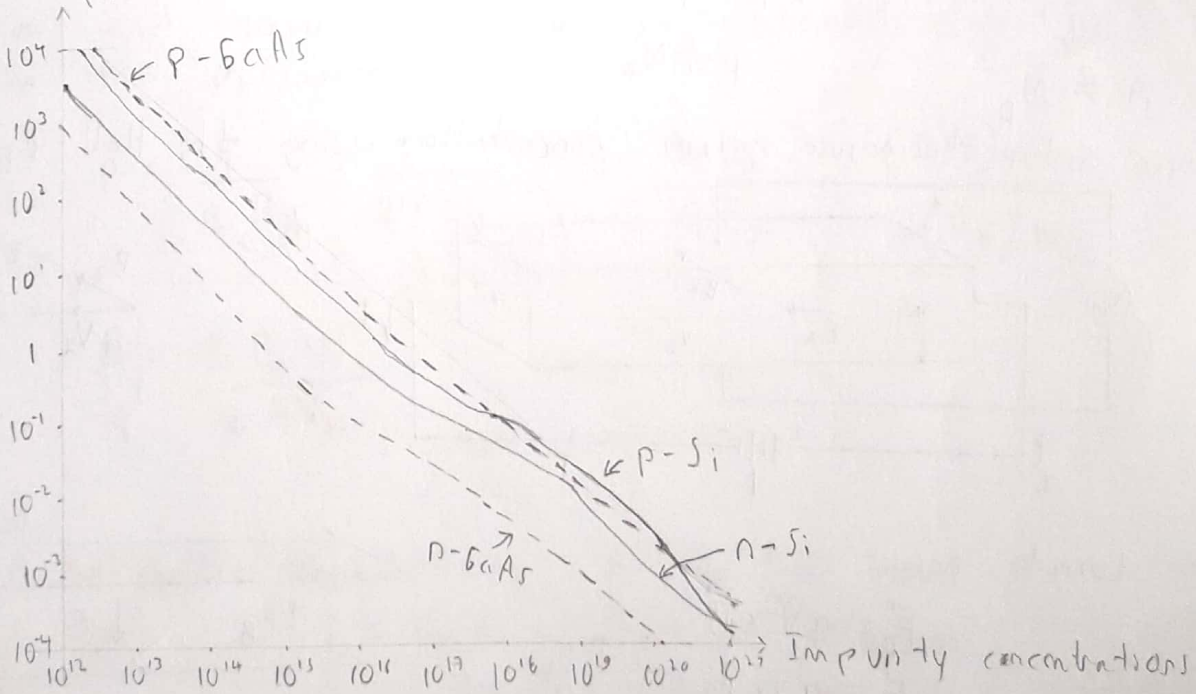
→ S = Probe spacing:



Four-point probe

→ $CF = CF\left(\frac{d}{s}\right)$ For $\frac{d}{s} > 20$: $CF = 4,54$

* At 300K



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EXAMPE 2

n-type silicon $\frac{10^{16} \text{ phosphorus}}{cm^3}$

At room temperature → All donors are ionized:

$$n = N_D = 10^{16} cm^{-3}$$

From figure $\rho(n\text{-}Si) = 5 \times 10^{-1} \Omega cm = 0{,}5 \Omega cm$.

From figure μ_n for silicon $\mu_n = 1300 \frac{cm^2}{Vs}$:

$$\rho = \frac{1}{q n \mu_n} = \frac{1}{(1{,}6 \times 10^{-19} C)\left(\frac{10^{16}}{cm^3}\right)\left(1300 \frac{cm^2}{Vs}\right)}$$

$$\rho = \frac{0{,}48}{\frac{A}{V cm}}$$

$$\rho = 0{,}48 \Omega\text{-}cm$$

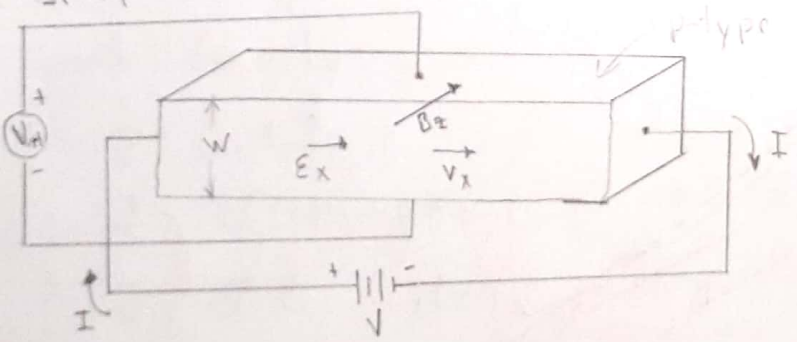
2.13 The Hall Effect

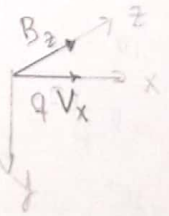
→ $n = n(T)$ → $p = p(T)$

↓

$n \neq N_D$ $\quad p \neq N_A$

* Basic setup to measure carrier concentration using the Hall effect:

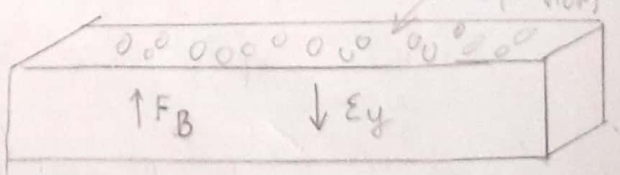




The Lorentz force:

$$\vec{F}_B = q\vec{V} \times \vec{B}$$

$$F_B = -q V_x B_z$$



$\uparrow F_y$

$\downarrow \mathcal{E}_y \rightarrow \downarrow F_H = \mathcal{E}_y q$

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There is no current flow along the y-direction in the steady state:

$$F_J + F_H = 0$$

$$-q V_x B_z + \mathcal{E}_y q = 0$$

$$q \mathcal{E}_y = q V_x B_z$$

$$\mathcal{E}_y = V_x B_z \leftarrow \text{Hall Field}$$

$$\uparrow F_y = 0$$

Hall-voltage:

$$V_H = \mathcal{E}_y W$$

$$\rightarrow \mathcal{E}_y = \left(\frac{J_p}{q p}\right) B_z$$

$$\rightarrow J_p = q p V_p$$

$$V_p = \frac{J_p}{q p}$$

$$\mathcal{E}_y = R_H J_p B_z \longrightarrow R_H = \frac{\mathcal{E}_y}{B_z J_p}$$

Hall coefficient:

$$R_H = \frac{1}{q p}$$

For an n-type semiconductor:

$$R_H = -\frac{1}{q n}$$

$$[R_H] = \frac{cm^3}{C}$$

Concentration and carrier type can be obtained directly from the Hall measurement

$$p = \frac{1}{q R_H} = \frac{1}{q\left(\frac{\mathcal{E}_y}{J_p B_z}\right)} = \frac{J_p B_z}{q \mathcal{E}_y} = \frac{(I/A) B_z}{q (V_H / W)}$$

$$p = \frac{I B_z W}{q A V_H}$$

EXAMPLE 3

$N_D = 10^{16} cm^{-3}$ $\quad W = 500 \mu m$ $\quad A = 2,5 \times 10^{-3} cm^2$

$V_H = ?$ $\quad I = 1 mA$ $\quad B_z = 10^{-4} Wb/cm^2$

$n = N_D$

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$$\rightarrow R_H = -\frac{1}{qn} = -\frac{1}{(1,6\times10^{-19}\,C)(10^{16}\,cm^{-3})}$$

$$R_H = -625\,cm^3/C$$

$$\rightarrow V_H = \mathcal{E}_y W = (R_H J_n B_z) W = R_H \left(\frac{I}{A}\right) B_z W$$

$$V_H = \left(-625\frac{cm^3}{C}\right)\frac{(10^{-3}A)}{(2,5\times10^{-3}cm^2)}(10^{-4}Wb/cm^2)(500\times10^{-4}cm)$$

$\vec{F} = q\vec{V}\times\vec{B}$

$\rightarrow Wb = [B][A]$

$$Wb = \frac{N}{C\,m/s}\,m^2$$

$$Wb = \frac{N\,m\,s}{C}$$

$$V_H = -1,25\times10^{-3}\frac{A\,Wb}{C}$$

$$V_H = -1,25\times10^{-3}\left(\frac{1}{s}\right)\left(\frac{N\,m\,s}{C}\right)$$

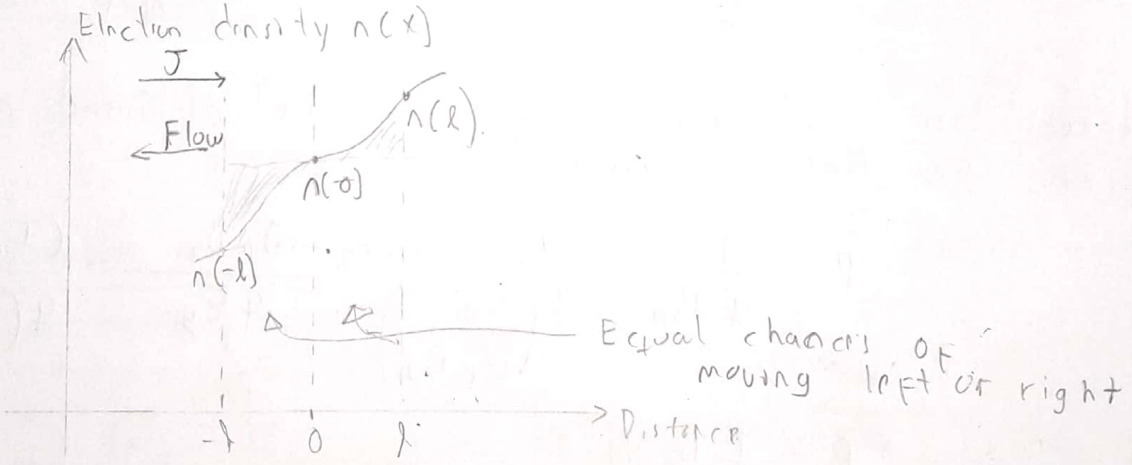
$$V_H = -1,25\times10^{-3}\frac{J}{C}$$

$$V_H = -1,25\,mV$$

2.2 CARRIER DIFUSSION.

2.2.1. Diffusion Process

Electron concentration versus distance: l = Mean free path:



Semiconductor is at uniform temperature:

⇩

Average thermal energy of electrons = constant

Because of finite temperature, the electrons have random thermal motion:

$l = v_{th}\tau_c$ τ_c: Mean free time

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The average rate of electron flow crossing plane $x=0$ from left:

$$[F] = \frac{\#}{\frac{[A]}{[t]}}$$

$$F_1 = \frac{\frac{1}{2}[n(-l)-n(0)]\,l}{\tau_c}$$

$$F_1 = \frac{1}{2}[n(-l)-n(0)]\,v_{th}$$

The average rate of electron flow crossing plane $x=0$ from right:

$$F_2 = \frac{\frac{1}{2}[n(+l)-n(0)]\,l}{\tau_c}$$

$$F_2 = \frac{1}{2}[n(+l)-n(0)]\,v_{th}$$

The net rate of carrier flow from left to right:

$$F = F_1 - F_2 = \frac{1}{2}v_{th}[n(-l)-n(+l)]$$

First two terms of a Taylor series expansion near to $a=0$, at $x=-l$ and $x=l$

$$F = \frac{1}{2}v_{th}\left\{\left[n(0) - l\frac{dn}{dx}\right] - \left[n(0) + l\frac{dn}{dx}\right]\right\}$$

Taylor series expansion

$$F(x) + \frac{F'(a)}{1!}(x-a) + \frac{F''(a)}{2!}(x-a)^2$$

$$F = -v_{th}\,l\frac{dn}{dx}$$

Diffusion coefficient or diffusivity: $D_n = v_{th}\,l$

$$F = -D_n\frac{dn}{dx}$$

Diffusion current:

$$J_n = (-q)F = (-q)\left(-D_n\frac{dn}{dx}\right) = qD_n\frac{dn}{dx}$$

$\frac{dn}{dx} > 0 \rightarrow$ F $\leftarrow$ $J_n \rightarrow$

For holes:

$$F = -D_p\frac{dp}{dx}$$

$$J_p = (q)F = -qD_p\frac{dp}{dx}$$

$\frac{dp}{dx} > 0 \rightarrow$ F $\leftarrow$ $J_p \rightarrow$

EXAMPLE 4

n-type semiconductor $T = 300\,K$

Electron concentration varies linearly:

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$$\frac{\Delta n}{\Delta x} = \frac{(10^{18}\,cm^{-3}) - (0,7 * 10^{18}\,cm^{-3})}{(0,1\,cm)} = 0,3 * 10^{18}\,cm^{-4}$$

$D_n = 22,5\,cm^2/s$

$$\rightarrow J_{n,diff} = q D_n \frac{dn}{dx} \approx q D_n \frac{\Delta n}{\Delta x}$$

$$= (1,6 * 10^{-19}\,C)(22,5\,cm^2/s)(0,3 * 10^{18}\,cm^{-4})$$

$$= 10,8\,\frac{A}{cm^2}$$

2.2.2 Einstein Relation

Using the theorem for the equipartition of energy for this one dimensional case

$$\frac{1}{2} m_n v_{th}^2 = \frac{1}{2} kT$$

$$v_{th}^2 = \frac{kT}{m_n}$$

$$\mu_n = \frac{q \tau_c}{m_n}$$

$$\tau_c = \frac{m_n \mu_n}{q}$$

$$\rightarrow D_n = v_{th} l = v_{th} (v_{th} \tau_c) = v_{th}^2 \left(\frac{m_n \mu_n}{q}\right)$$

$$D_n = v_{th}^2 \left(\frac{m_n \mu_n}{q}\right)$$

$$\boxed{D_n = \left(\frac{kT}{q}\right) \mu_n}$$

Einstein relation

At $T = 300K$

$$\frac{kT}{q} = 25,9\,mV$$

Examples:

$\rightarrow D_n (\mu_n = 1000\,cm^2/Vs) =$

$= (25,9\,mV)(1000\,cm^2/Vs)$

$= 259\,cm^2/s$ ← Si

$\rightarrow D_p (20\,cm^2/Vs) =$ $N_D = 10^{14}\,cm^{-3}$

$= (25,9\,mV)(20\,cm^2/Vs)$

$= 0,518\,cm^2/s$

↓ (p) GaAs

$N_A = 10^{16}\,cm^{-3}$

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EXAMPLE 5

holes are injected into a homogeneous n-type

$\mathcal{E} = 50 V/cm \longrightarrow \ell = 1 cm \quad \tau_t = 100 \mu s \quad V_p = ? \quad D_p = ? \quad T = 300 K$

$\rightarrow V_p = \frac{\ell}{\tau_t} = \frac{(1 cm)}{(100 \mu s)} = 10^4 cm/s.$

$\rightarrow V_p = \mu_p \mathcal{E}$

$\rightarrow \mu_p = \frac{V_p}{\mathcal{E}} = \frac{(10^4 cm/s)}{(50 V/cm)} = 200 \frac{cm^2}{V-s}$

$\rightarrow D_p = \left(\frac{kT}{q}\right)\mu_p$

$= (0,0259 V)\left(200 \frac{cm^2}{V-s}\right)$

$D_p = 5,18 cm^2/s$

2.2.3 Current Density Equations

The total current density at any point:

$$J_n = J_{n,drift} + J_{n,diff}$$

$$\boxed{J_n = q n \mu_n \mathcal{E} + q D_n \frac{dn}{dx}}$$

Hole current density:

$$J_p = J_{p,drift} + J_{p,diff}$$

$$\boxed{J_p = q p \mu_p \mathcal{E} - q D_p \frac{dp}{dx}}$$

Under low electric fields

Total conduction current density:

$$\boxed{J_{cond} = J_n + J_p}$$

2.3 GENERATION AND RECOMBINATION PROCESSES

In thermal equilibrium: $pn = n_i^2$

If excess carriers are introduced: $pn > n_i^2$ → Nonequilibrium situation

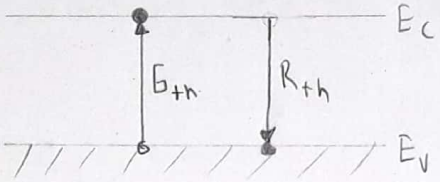
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2.3.1 Direct Recombination (GaAs)

Generation rate: $[G] = \dfrac{\left(\dfrac{\#\ \text{electron-hole pairs generated}}{cm^3}\right)}{\text{second}}$

$$[G] = cm^{-3}\,s^{-1}$$

* Direct generation and recombination at thermal equilibrium:



Under thermal equilibrium conditions:

$$G_{th} = R_{th}$$

$\Downarrow$

Carrier concentration remain constant

$\Downarrow$

$$pn = n_i^2$$

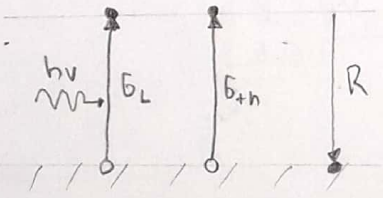
For a n-type semiconductor:

$$G_{th} = R_{th} = \beta\, n_{no}\, p_{no}$$

β: proportionality constant

$$[\beta] = \frac{cm^3}{s}$$

* Direct generation and recombination under illumination:



- G_L: Rate of electron-hole pairs when we shine a light

$\rightarrow R = \beta np$

For a n-type semiconductor:

$$R = \beta n_n p_n = \beta (n_{no} + \Delta n)(p_{no} + \Delta p)$$

And:

$$G = G_L + G_{th}$$ (G_{th}: At thermal equilibrium)

Excess carrier concentrations:

$$\Delta n = n_n - n_{no}$$

$$\Delta p = p_n - p_{no}$$

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To maintain overall charge neutrality:

$$\Delta n = \Delta p$$

The net rate of change of hole concentration:

$$\frac{dp_n}{dt} = G - R$$

$$\frac{dp_n}{dt} = G_L + G_{th} - R$$

In steady state: $G - R = 0 = \frac{dp_n}{dt}$

$$0 = G_L + G_{th} - R$$

$$G_L + G_{th} = R$$

$$G_L = R - G_{th}$$

Net recombination rate: $U = G_L$

$$U = R - G_{th}$$

$$U = [\beta (n_{n_0} + \Delta n)(p_{n_0} + \Delta p)] - (\beta n_{n_0} p_{n_0})$$

$$U = \beta [(n_{n_0} + \Delta p)(p_{n_0} + \Delta p) - n_{n_0} p_{n_0}]$$

$$U = \beta [\Delta p (p_{n_0} + n_{n_0}) + \Delta p^2]$$

$$U = \beta (n_{n_0} + p_{n_0} + \Delta p) \Delta p$$

For low-level injection: $\Delta p << n_{n_0}$ Beside: $p_{n_0} << n_{n_0}$:

$$U \cong \beta n_{n_0} \Delta p$$

$$U = \frac{p_n - p_{n_0}}{\frac{1}{\beta n_{n_0}}}$$

Lifetime: $\tau_p = \frac{1}{\beta n_{n_0}}$

$$U = \frac{p_n - p_{n_0}}{\tau_p}$$

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In thermal equilibrium and steady state:

$$p_n - p_{n_0} = 0$$

$$\downarrow$$

$$U = 0$$

* Decay of photoexcited carriers:

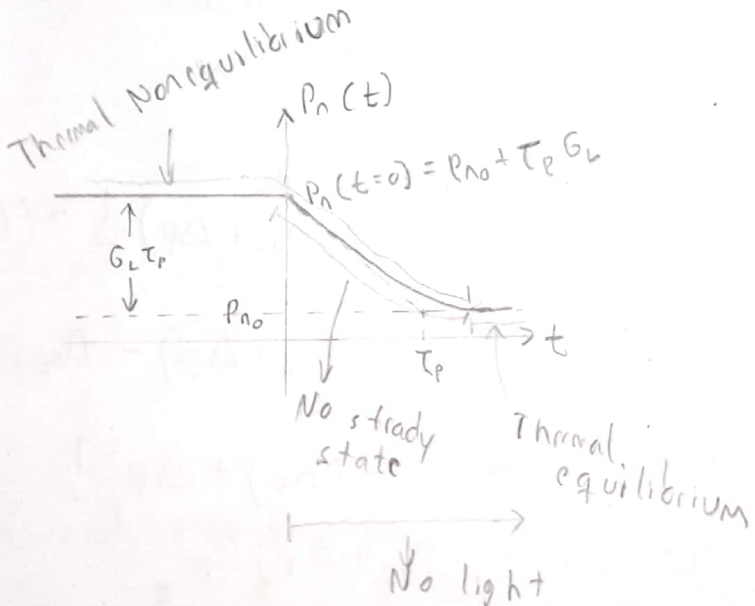
In steady state:

$$G_L = U = \frac{p_n - p_{n_0}}{\tau_p} = \frac{\Delta p_n}{\tau_p}$$

$$\Delta p = \Delta n = G_L \tau_p$$

If at an arbitrary time, $t = 0$, the light is suddenly turned OFF:

$$p_n(t=0) = p_{n_0} + \tau_p G_L$$



The time-dependent expression (No light): $G_L = 0$

$$\frac{dp_n}{dt} = \cancel{G_L}^{0} + G_{th} - R$$

$$\frac{dp_n}{dt} = -U$$

$$\frac{dp_n}{dt} = \frac{p_{n_0} - p_n}{\tau_p}$$

For low level injection

$$-U = -(R - G_{th})$$

$$-U = -\left(\frac{p_n - p_{n_0}}{\tau_p}\right)$$

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$$\int_{P_n(t=0)}^{P_n} \frac{dP_n}{P_{n0} - P_n} = \int_0^t \frac{dt}{\tau_p}$$

$$-Ln\,|P_{n0} - P_n|\,\Big|_{P_n(t=0)}^{P_n} = \frac{t}{\tau_p}$$

$P_n > P_{n0}$ ← Forever

$$-Ln\,(P_n - P_{n0})\,\Big|_{P_n(t=0)}^{P_n} = \frac{t}{\tau_p}$$

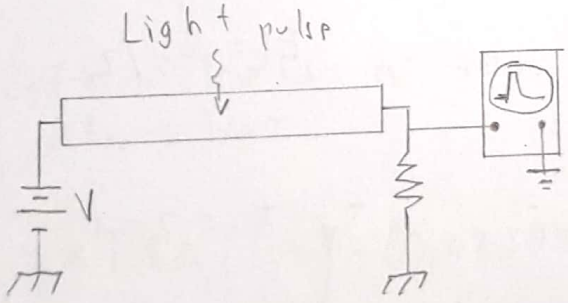
$$Ln\,(P_n - P_{n0})\,\Big|_{P_n(t=0)}^{P_n} = -\frac{t}{\tau_p}$$

$$Ln\left(\frac{P_n - P_{n0}}{P_n(t=0) - P_{n0}}\right) = -\frac{t}{\tau_p}$$

$$\frac{P_n - P_{n0}}{(P_{n0} + \tau_p \sigma_L) - P_{n0}} = e^{-t/\tau_p}$$

$$P_n = P_{n0} + \tau_p \sigma_L e^{-t/\tau_p}$$

Schematic setup to measure minority carrier lifetime! (photoconductivity method):



(Constant current:

Increasing Voltage ← Decay conductivity

↓ Measuring carrier lifetime

↑ Decay excess carriers

↑ Without light

EXAMPLE 6

Si : $n_{n0} = 10^{14}\ cm^{-3}$ $\quad \sigma_L = \frac{10^{17}\ \text{electron-hole pairs}}{cm^3\ \mu s}$ $\quad \tau_n = \tau_p = 2\mu s$

$\Delta P_n = ?$ ← Change in the minority carrier concentration

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Before illumination:

For silicon: $n_i (T = 27°C) = 9{,}65 * 10^9 \, cm^{-3}$

$$p_{n0} = \frac{n_i^2}{n_{n0}} = \frac{(9{,}65 * 10^9 \, cm^{-3})^2}{10^{14} \, cm^{-3}} = 9{,}31 * 10^5 \, cm^{-3}$$

After illumination:

$$\Delta p_n = \tau_p G_L = (2\mu s)\left(\frac{10^{13} \, cm^{-3}}{\mu s}\right) = 2 * 10^{13} \, cm^{-3}$$

$$p_n = p_{n0} + \tau_p G_L = 9{,}31 * 10^5 \, cm^{-3} + 2 * 10^{13} \, cm^{-3} \approx 2 * 10^{13} \, cm^{-3}$$

$$p_n \approx 2 * 10^{13} \, cm^{-3}$$

2.3.2 Quasi-Fermi Level

Under illumination → Excess carriers → Electron and hole concentrations are higher than those in equilibrium state:

$$p_n > n_i^2$$

Quasi-Fermi levels: E_{Fn} / E_{Fp}:

→ $n = n_i e^{(E_{Fn} - E_i)/kT}$

→ $p = n_i e^{(E_i - E_{Fp})/kT}$

E_F ctte ← only thermal equilibrium.

→ E_i: Endrinsic Fermi Level:



$E_i = E_g / 2$

EXAMPLE 7

GaAs sample with $n_{n0} = 10^{16} \, cm^{-3}$ $\tau_p = \tau_n = 2ns$

↓

n type

$$G_L = 10^{13}\left(\frac{\text{electrons - hole pairs}}{cm^3}\right)/\mu s$$

$$G_L = 10^{13} \frac{cm^{-3}}{\mu s}$$

For GaAs: $n_i = 2{,}25 * 10^6 \, cm^{-3}$

$E_F = E_{Fn} = ?$ $E_{Fp} = ?$ } At room temperature

(Before illumination:

→ $n_{n0} = 10^{16} \, cm^{-3}$

→ $p_{n0} = \frac{n_i^2}{n_{n0}} = \frac{(2{,}25 * 10^6 \, cm^{-3})^2}{(10^{16} \, cm^{-3})} = 5{,}06 * 10^{-4} \, cm^{-3}$

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$$p_{n_0} = \frac{5,06 * 10^{-4}}{cm^3} \left(\frac{10^2 cm}{1 m} \right)^3 = 5,06 * 10^2 \, m^{-3}$$

Intrinsic Fermi Level: $E_i = \frac{E_g}{2} = \frac{1,42 eV}{2} = 0,71 eV$ ← For GaAs

After ; Illumination :

$$\rightarrow G_L = \frac{n_n - n_{n_0}}{\tau_n}$$

$$n_n = n_{n_0} + \tau_n G_L = (10^{16} cm^{-3}) + (2 ns)\left(10^{13} \frac{cm^{-3}}{ns}\right)$$

$$= 10^{16} cm^{-3} + 2 * 10^{-3} * 10^{13} cm^{-3}$$

$$n_n \approx n_{n_0} = 10^{16} cm^{-3} \leftarrow \text{Low injection}$$

$$\rightarrow G_L = \frac{p_n - p_{n_0}}{\tau_p}$$

$$p_n = p_{n_0} + \tau_p G_L$$

$$p_n = (5,06 * 10^{-4} cm^{-3}) + (2 ns)\left(10^{13} \frac{cm^{-3}}{ns}\right)$$

$$p_n = 5,06 * 10^{-4} cm^{-3} + 2 * 10^{10} cm^{-3}$$

$$p_n \approx \tau_p G_L = 2 * 10^{10} cm^{-3}$$

The quasi-Fermi levels at room temprature:

$$\rightarrow n_n = n_i \, e^{(E_{Fn} - E_i)/kT}$$

$$E_{Fn} - E_i = kT \ln\left(\frac{n_n}{n_i}\right) = (0,0259 eV) \ln\left(\frac{10^{16} cm^{-3}}{2,25 * 10^6 cm^{-3}}\right) = 0,575 eV$$

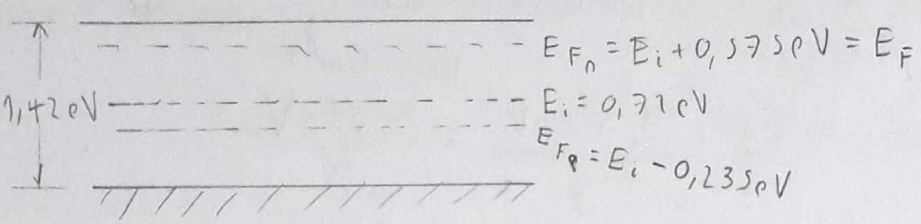
$$E_{Fn} = 0,575 eV + E_i$$

$$\rightarrow p_n = n_i \, e^{(E_i - E_{Fp})/kT}$$

$$E_i - E_{Fp} = kT \ln\left(\frac{p_n}{n_i}\right) = (0,0259 eV) \ln\left(\frac{2 * 10^{10} cm^{-3}}{2,25 * 10^6 cm^{-3}}\right) = 0,235 eV$$

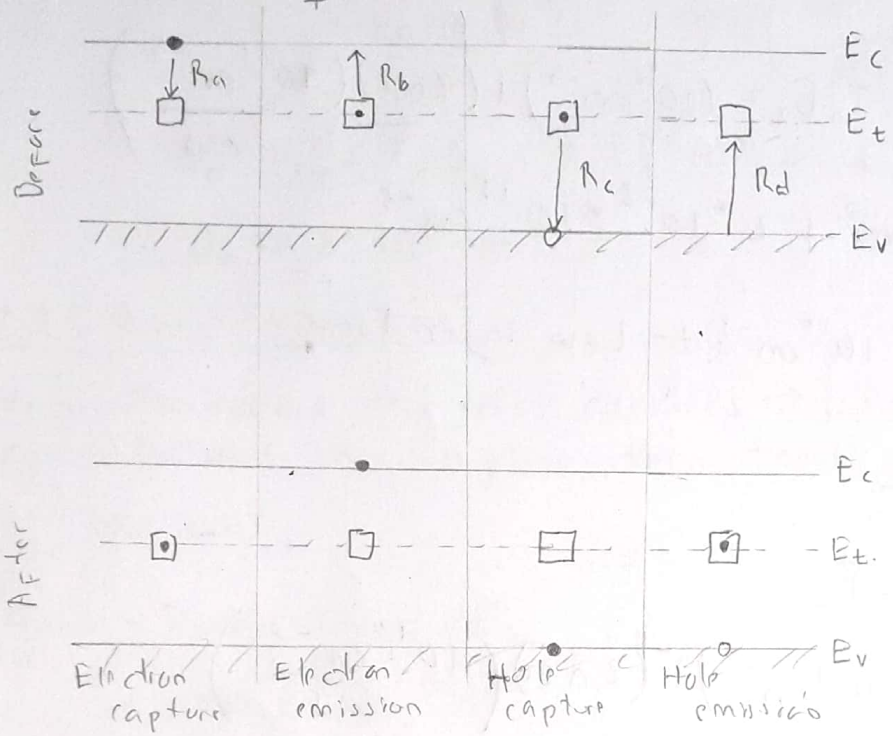
$$E_{Fp} = E_i - 0,235 eV.$$

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2.3.3 Indirect Recombination

At thermal equilibrium:



→ Intermediate-level states = Recombinations centers

$E_t \rightarrow \square \rightarrow$ Neutral

$E_t \rightarrow \boxdot \rightarrow$ Negative

Recombination rate:

$$U = \frac{V_{th}\sigma_n\sigma_p N_t (p_n n_n - n_i^2)}{\sigma_p [p_n + n_i e^{(E_i - E_t)/kT}] + \sigma_n [n_n + n_i e^{(E_t - E_i)kT}]}$$

$[U] = \frac{cm^{-3}}{s}$

V_{th}: Thermal velocity:

$$\frac{1}{2} m_n V_{th} = \frac{3}{2} kT \quad \text{or} \quad \frac{1}{2} m_p V_{th} = \frac{3}{2} kT$$

N_t: Concentration of the recombination center in the semiconductor

$[N_t] = cm^{-3}$

σ_n = Electrons capture cross section

σ_p = Capture cross-section of holes

$[\sigma] = cm^2$

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Assuming equal electron and hole capture cross sections:

$\rightarrow \sigma_n = \sigma_p = \sigma_0$

$$\rightarrow U = \frac{V_{th}\,\sigma_0\,\sigma_0\,N_t\,(p_n n_n - n_i^2)}{\sigma_0\,[p_n + n_i\, e^{(E_i - E_t)/kT}] + \sigma_0\,[n_n + n_i\, e^{(E_t - E_i)/kT}]}$$

$$U = V_{th}\,\sigma_0\,N_t\,\frac{(p_n n_n - n_i^2)}{p_n + n_n + n_i * 2\left[\frac{e^{-(E_t - E_i)/kT} + e^{(E_t - E_i)/kT}}{2}\right]}$$

$$U = V_{th}\,\sigma_0\,N_t\,\frac{(p_n n_n - n_i^2)}{p_n + n_n + 2\,n_i \cosh\left(\frac{E_t - E_i}{kT}\right)}$$

Under a low-injection condition in an n-type semiconductor $n_n >> p_n$:

$$U \approx V_{th}\,\sigma_0\,N_t\,\frac{\left(\frac{p_n n_n}{n_n} - \frac{n_i^2}{n_n}\right)}{\frac{p_n}{n_n}^{\to 0} + \frac{n_n}{n_n} + \frac{2n_i}{n_n}\cosh\left(\frac{E_t - E_i}{kT}\right)}$$

$$U = V_{th}\,\sigma_0\,N_t\,\frac{(p_n - p_{n0})}{1 + \frac{2n_i}{n_n}\cosh\left(\frac{E_t - E_i}{kT}\right)}$$

$$\rightarrow \tau_p = \frac{1 + \frac{2n_i}{n_n}\cosh\left(\frac{E_t - E_i}{kT}\right)}{V_{th}\,\sigma_0\,N_t}$$

$$[\tau_p] = \frac{1}{\frac{cm}{s}\,cm^2\,cm^{-3}} = s$$

$$\rightarrow [\sigma] = \frac{[U]}{[V_{th}][N][P]}$$

$$[\sigma] = \frac{\frac{cm^{-3}}{s}}{\frac{cm}{s}\,cm^{-3}\,cm^{-3}}$$

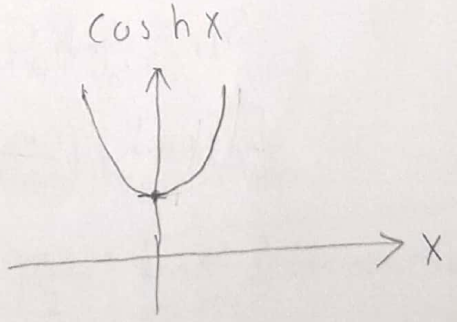
$$[\sigma] = cm^2$$

$\rightarrow n_n \approx n_{n0}$

So;

$n_i^2 = p_{n0}\,n_{n0}$

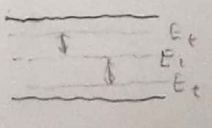
$p_{n0} \approx \frac{n_i^2}{n_n}$



So:

$$\rightarrow U = \frac{p_n - p_{n0}}{\tau_p} \quad \text{and} \quad \tau_p = \tau_p(E_t)$$

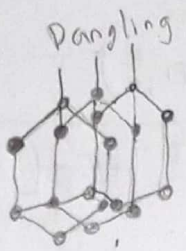
$\tau_p \propto |E_t - E_i|$ Proportional to:



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2.3.4 Surface Recombination

Bonds at a clean semiconductor surface:



→ Bonds are anisotropic

↘ These energy states, called surface states

For a low-injection condition ($n_n >> p_n$) and $n_{ns} = n_n$ (at the surface):

Total number of carriers recombining at the surface:

$$U_s = V_{th}\,\sigma_p\,N_{st}\,(p_s - p_{no}) \qquad [U_s] = \left(\frac{1}{cm^2}\right)/s$$

$$[U_s] = \frac{cm^{-2}}{s}$$

→ Hole concentrations at the surface: p_s $\quad [p_s] = cm^{-3}$

→ N_{st}: Recombination center density in the surface region.

$$[N_{st}] = cm^{-2}$$

Low injection surface recombination velocity S_{lr}:

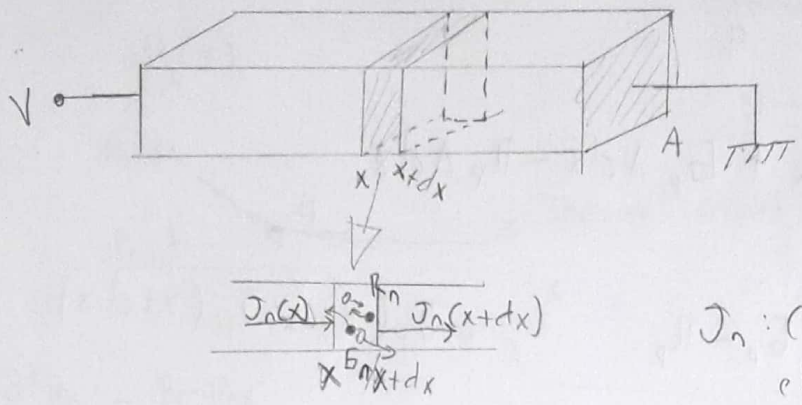
$$S_{lr} = V_{th}\,\sigma_p\,N_{st}$$

$$[S_{lr}] = \left(\frac{cm}{s}\right)\left(\cancel{cm^{-2}}\right)\left(\cancel{cm^{-2}}\right)$$

$$[S_{lr}] = \frac{cm}{s}$$

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2.4 CONTINUITY EQUATION



J_n: Current density due to electrons

The overall rate of change in the number of electrons in the slice:

$$\frac{\partial(\#\text{ of electrons})}{\partial t} = \#\text{ electrons flowing into at } x - \#\text{ electrons flowing out at } x+dx + \#\text{ electrons are generated} - \#\text{ electrons are recombined.}$$

$$\frac{\partial n}{\partial t} A dx = \frac{J_n(x)A}{-q} - \frac{J_n(x+dx)A}{-q} + G_n A dx - R_n A dx$$

$\rightarrow A dx = V$ of the slice

$$\frac{\partial n}{\partial t} A dx = \frac{A}{q}\left[J_n(x+dx) - J_n(x)\right] + \left[G_n - R_n\right] A dx$$

Taylor series yields:

$$J_n(x+dx) = J_n(x) + \frac{\partial J_n}{\partial x} dx + \ldots$$

$$\frac{\partial n}{\partial t} dx = \frac{1}{q}\left[\frac{\partial J_n}{\partial x} dx\right] + \left(G_n - R_n\right) dx$$

Continuity equation for electrons:

$$\frac{\partial n}{\partial t} = +\frac{1}{q}\frac{\partial J_n}{\partial x} + \left(G_n - R_n\right)$$

$\rightarrow J_n(x+dx) > J_n(x)$

For the one-dimensional case under low-injection condition:

$$\frac{\partial n_p}{\partial t} = \frac{1}{q}\frac{\partial}{\partial x}\left(q\mu_n n_p \mathcal{E} + q D_n \frac{dn_p}{dx}\right) + \left(G_{Ln} + G_{th}\right) - R_n$$

Net recombination rate:

$$U = R_n - G_{th_n} = \frac{n_p - n_{p_0}}{\tau_n}$$

$$\boxed{\frac{\partial n_p}{\partial t} = n_p \mu_n \frac{\partial \mathcal{E}}{\partial x} + \mathcal{E}\mu_n \frac{\partial n_p}{\partial x} + D_n \frac{\partial^2 n_p}{\partial x^2} + G_{Ln} - \frac{n_p - n_{p_0}}{\tau_n}}$$

light

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For holes:

$$\frac{\partial p}{\partial t} A dx = \frac{J_p(x)}{q} A - \frac{J_p(x+dx)}{q} A + G_p A dx - R_p A dx$$

$$\frac{\partial p}{\partial t} A dx = -\frac{A}{q} \frac{\partial J_p}{\partial x} dx + G_p A dx - R_p A dx$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q} \frac{\partial J_p}{\partial x} + G_p - R_p$$

$J_p(x) \xrightarrow{p} J_p(x+dx)$

For the one-dimensional case under low-injection condition:

$$\frac{\partial p_n}{\partial t} = -\frac{1}{q} \frac{\partial}{\partial x}\left(q \mu_p p_n \mathcal{E} - q D_p \frac{dn_p}{dx}\right) + (G_{Lp} + G_{thp}) - R_p$$

$$\boxed{\frac{\partial p_n}{\partial t} = -p_n \mu_p \frac{\partial \mathcal{E}}{\partial x} - \mathcal{E} \mu_p \frac{\partial p_n}{\partial x} + D_p \frac{\partial^2 p_n}{\partial x^2} + G_{Lp} - \frac{p_n - p_{n0}}{\tau_p}}$$

Net recombination rate:

$U = R_p - G_{thp}$

$U = \frac{p_n - p_{n0}}{\tau_p}$

Poisson's Equation:

$$\boxed{\frac{d\mathcal{E}}{dx} = \frac{\rho_s}{\varepsilon_s}}$$

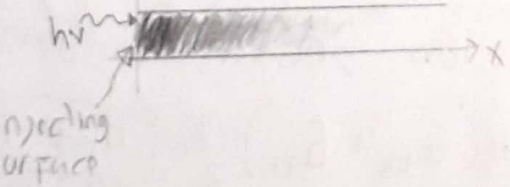
Space charge density:

$$\rho_s = q(p - n + N_D^+ - N_A^-)$$

ionized impurity concentration

2.4.1 Steady-state injection from One side.

An n-type semiconductor:



$\rightarrow \mathcal{E} = 0$

Inside the semiconductor:

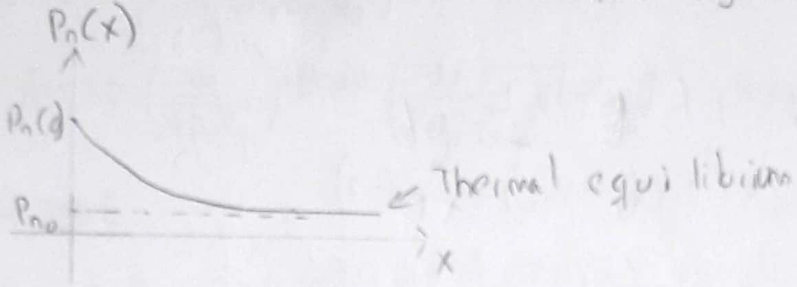
$$\frac{\partial p_n}{\partial t}^{0} = -p_n \mu_p \frac{\partial \mathcal{E}}{\partial x}^{0} - \mu_p \mathcal{E} \frac{\partial p_n}{\partial x} + D_p \frac{\partial^2 p_n}{\partial x^2} + G_{Lp}^{0} - \frac{p_n - p_{n0}}{\tau_p}$$

$\rightarrow$ study state

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Boundary conditions: $\rightarrow P_n(x=0) = P_n(0)$ $\rightarrow P_n(x \to \infty) = P_{n_0}$

↑ Due light.



$$\rightarrow 0 = D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n - P_{n_0}}{\tau_p}$$

$$-\frac{P_{n_0}}{D_p \tau_p} = \frac{\partial^2 P_n}{\partial x^2} + \frac{(0)\partial P_n}{D_p \partial x}^{\to 0} - \frac{P_n}{D_p \tau_p}$$

For homogeneus equation:

$$0 = r^2 - \frac{1}{D_p \tau_p} \qquad \rightarrow P_n = A\,exp^{x/\sqrt{D_p \tau_p}} + B\,exp^{-x/\sqrt{D_p \tau_p}}$$

$$r = \pm \frac{1}{\sqrt{D_p \tau_p}}$$

For no homogeneus equation:

$$\rightarrow P_n = P_{n_0}$$

Thus, the complete solution is:

$$\rightarrow P_n = P_{n_0} + A\,exp^{x/\sqrt{D_p \tau_p}} + B\,exp^{-x/\sqrt{D_p \tau_p}}$$

Through the boundary conditions:

$\rightarrow P_n(x=0) = P_n(0)$

$\rightarrow P_n(0) = P_{n_0} + A + B$

$P_n(0) = P_{n_0} + B$

$B = P_n(0) - P_{n_0}$

$\rightarrow P_n(x \to \infty) = P_{n_0}$

$\rightarrow P_n(x \to \infty) = P_{n_0} + A \lim_{x \to \infty} e^{x/\sqrt{D_p \tau_p}} + B \lim_{x \to \infty} e^{-x/\sqrt{D_p \tau_p}}$ (the second limit $\to 0$)

Only: $A = 0$

$P_n(x \to \infty) = P_{n_0}$

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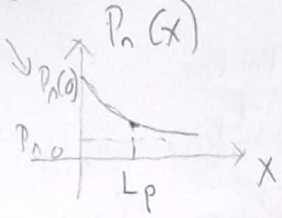
So, the solution is:

$$P_n(x) = P_{n0} + (P_n(0) - P_{n0}) \exp\left(-\frac{x}{\sqrt{D_p \tau_p}}\right).$$

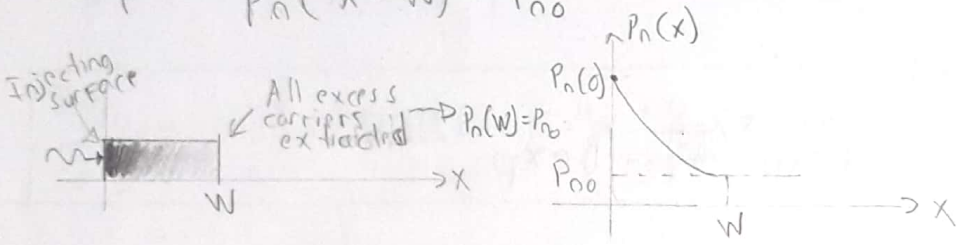
→ Diffusion length: $L_p = \sqrt{D_p \tau_p}$:

$$P_n(L_p) = P_{n0} + (P_n(0) - P_{n0}) \exp(-1)$$

$$P_n(L_p) = P_{n0} + (P_n(0) - P_{n0}) 36{,}8\%$$



* If $P_n(x = W) = P_{n0}$



$$\frac{\partial P_n}{\partial t} = -P_n \mu_p \frac{\partial \mathcal{E}}{\partial x} - \mu_p \mathcal{E} \frac{\partial P_n}{\partial x} + D_p \frac{\partial^2 P_n}{\partial x} + G_{L_p} - \frac{P_n - P_{n0}}{\tau_p}$$

At steady state:

$$0 = D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n - P_{n0}}{\tau_p}$$

The boundary conditions:

→ $P_n(x = 0) = P_n(0)$

↑ Due the light

→ $P_n(W) = P_{n0}$

The complete solution look like before:

→ $P_n = P_{n0} + A \exp\left(\frac{x}{\sqrt{D_p \tau_p}}\right) + B \exp\left(-\frac{x}{\sqrt{D_p \tau_p}}\right)$

Applicated the boundary conditions:

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$$\rightarrow P_n(0) = P_{no} + A + B$$

$$\rightarrow P_n(W) = \cancel{P_{no}} = P_{no} + A\exp\left(\frac{W}{\sqrt{D_p \tau_p}}\right) + B\exp\left(-\frac{W}{\sqrt{D_p \tau_p}}\right)$$

$$0 = A\exp\left(\frac{W}{\sqrt{D_p \tau_p}}\right) + B\exp\left(\frac{-W}{\sqrt{D_p \tau_p}}\right)$$

$$\rightarrow P_n(0) = P_{no} + A + B \qquad (+)$$

$$-P_n(0)\exp\left(\frac{W}{L_p}\right) = -P_{no}\exp\left(\frac{W}{L_p}\right) - A\exp\left(\frac{W}{L_p}\right) - B\exp\left(\frac{W}{L_p}\right)$$

$$\rightarrow -P_n(0)\exp\left(\frac{W}{L_p}\right) = -P_{no}\exp\left(\frac{W}{L_p}\right) + B\left[\exp\left(-\frac{W}{L_p}\right) - \exp\left(\frac{W}{L_p}\right)\right]$$

$$\frac{[P_{no} - P_n(0)]\exp\left(\frac{W}{L_p}\right)}{\exp\left(-\frac{W}{L_p}\right) - \exp\left(\frac{W}{L_p}\right)} = B = \frac{[P_n(0) - P_{no}]\exp\left(\frac{W}{L_p}\right)}{2\sinh\left(\frac{W}{L_p}\right)}$$

On the other hand:

$$P_n(0) = P_{no} + A + \left\{\frac{[P_n(0) - P_{no}]\, e^{W/L_p}}{2\sinh(W/L_p)}\right\}$$

$$A = P_n(0) - P_{no} - \frac{[P_n(0) - P_{no}]\, e^{W/L_p}}{2\sinh(W/L_p)}$$

$$A = (P_n(0) - P_{no})\left[1 - \frac{e^{W/L_p}}{2\sinh(W/L_p)}\right] = (P_n(0) - P_{no})\left[\frac{e^{W/L_p} + e^{-W/L_p} - 1}{e^{-W/L_p} - 1}\right]$$

Thus:

$$P_n = P_{no} + (P_n(0) - P_{no})\left[\frac{e^{W/L_p} + e^{-W/L_p} - 1}{e^{-W/L_p} - 1}\right] e^{x/L_p} + \frac{[P_{no} - P_n(0)]\, e^{W/L_p}}{e^{-W/L_p} - 1} e^{-x/L_p}$$

$$P_n = P_{no} + \frac{[P_n(0) - P_{no}]}{e^{-W/L_p} - 1}\left[\left(e^{W/L_p} + e^{-W/L_p} - 1\right)e^{x/L_p} - e^{W/L_p}\, e^{-x/L_p}\right]$$

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$$P_n = P_{no} + \frac{[P_n(0) - P_{no}]}{e^{-W/L_p} - 1}\left[e^{W/L_p}e^{x/L_p} + e^{-W/L_p}e^{+x/L_p} - e^{x/L_p} - e^{W/L_p}e^{-x/L_p}\right]$$

$$P_n = P_{no} + \frac{[P_n(0) - P_{no}]}{e^{-W/L_p} - 1}\left\{e^{x/L_p}\left(e^{W/L_p} - 1\right) + \left[e^{(W-x)/L_p} - e^{-(W-x)/L_p}\right]\right\}$$

On the other hand:

$$\rightarrow P_n(0) = P_{no} + A + B$$

$$-P_n(0)\exp\left(-\frac{W}{L_p}\right) = -P_{no}\exp\left(-\frac{W}{L_p}\right) - A\exp\left(-\frac{W}{L_p}\right) - B\exp\left(-\frac{W}{L_p}\right)$$

Too:

$$0 = A\exp\left(W/L_p\right) + B\exp\left(-W/L_p\right)$$

So:

$$\rightarrow -P_n(0)\exp\left(-W/L_p\right) = -P_{no}\exp\left(-\frac{W}{L_p}\right) + A\left[\exp\left(\frac{W}{L_p}\right) - \exp\left(-\frac{W}{L_p}\right)\right]$$

$$A = \frac{\exp\left(\frac{W}{L_p}\right)\left[-P_n(0) + P_{no}\right]}{2\sinh\left(\frac{W}{L_p}\right)} = \frac{-\left[P_n(0) - P_{no}\right]\exp\left(-\frac{W}{L_p}\right)}{2\sinh\left(\frac{W}{L_p}\right)}$$

Thus:

$$P_n = P_{no} + Ae^{x/L_p} + Be^{-x/L_p}$$

$$P_n = P_{no} + \left\{\frac{-\left[P_n(0) - P_{no}\right]\exp\left(-W/L_p\right)}{2\sinh\left(\frac{W}{L_p}\right)}\right\}e^{x/L_p}$$

$$+ \left\{\frac{\left[P_n(0) - P_{no}\right]e^{W/L_p}}{2\sinh\left(W/L_p\right)}\right\}e^{-x/L_p}$$

$$P_n = P_{no} + \left\{\frac{P_n(0) - P_{no}}{2\sinh\left(W/L_p\right)}\right\}\left[e^{(W-x)/L_p} - e^{-(W-x)/L_p}\right]$$

$$P_n = P_{no} + \left[P_n(0) - P_{no}\right]\left[\frac{\sinh\left(\frac{W-x}{L_p}\right)}{\sinh\left(W/L_p\right)}\right]$$

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Current density (of holes) at $x = W$:

$$J_p(W) = q\mu_p p\mathcal{E} - q D_p \frac{dp_n}{dx}\Big|_{x=W}$$

$\mathcal{E} = 0$

$$\frac{d}{dx}\sinh x = \cosh x$$

$$J_p(W) = -q D_p \frac{d}{dx}\left\{ p_{no} + [p_n(0) - p_{no}]\left[\frac{\sinh\left(\frac{W-x}{L_p}\right)}{\sinh(W/L_p)}\right]\right\}\Big|_{x=W}$$

$$J_p(W) = -q D_p [p_n(0) - p_{no}] \frac{(-1/L_p)\cosh\left(\frac{W-x}{L_p}\right)}{\sinh(W/L_p)}\Big|_{x=W}$$

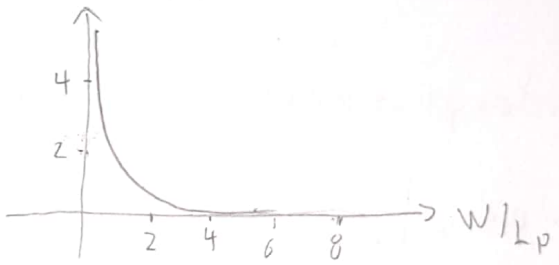
$\cosh\theta = 1$

$$J_p(W) = q\frac{D_p}{L_p}[p_n(0) - p_{no}]\frac{1}{\sinh(W/L_p)}$$

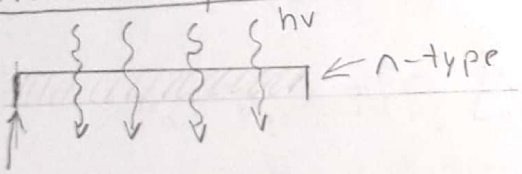
Then:

$$J_p(W) \propto \frac{1}{\sinh\left(\frac{W}{L_p}\right)}$$

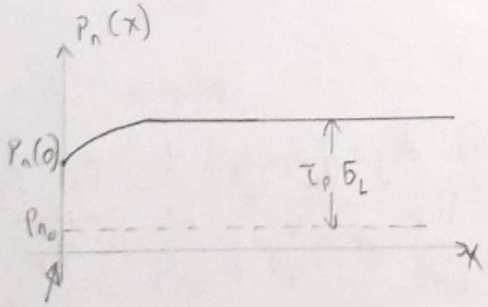
$1/\sinh(W/L_p)$



2.42. Minoritary Carriers at the Surface



Surface recombination at $x = 0$



Surface recombination at $x = 0$

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Hole current density flowing into the surface from the bulk (to left)

$$J_{ps} = -q\left(\text{Total number of carriers recombining at the surface}\right)$$

$$J_{ps} = -q\, U_s$$

$$[U_s] = \frac{cm^2}{s}$$

$$J_{ps} = -q\left(\text{Low injection surface recombination velocity}\right)(P_s - P_{no})$$

$$J_{ps} = -q\, S_{ar}\,(P_n(0) - P_{no})$$

$$S_{ar} = V_{th}\,\sigma_p\,N_{st}$$

N_{st}: Recombination center density in the surface

Gradient of hole concentration yields a diffusion current density:

$$J_{pdiff} = -q\, D_p \frac{dp_n}{dx}$$ (To left)

The boundary condition at $x = 0$

$$J_{pdiff}\big|_{x=0} = J_{ps}$$

$$-q\, D_p \frac{dp_n}{dx}\Big|_{x=0} = -q\, S_{ar}\,[P_n(0) - P_{no}]$$

$$D_p \frac{dp_n}{dx}\Big|_{x=0} = S_{ar}\,[P_n(0) - P_{no}]$$

The boundary condition at $x \to \infty$:

$$P_n\big|_{x\to\infty} = P_{no} + \tau_p\, G_L$$

At steady state: (It is constant)

$$\frac{\partial P_n}{\partial t} = 0 = -P_n \mu_p \frac{\partial \mathcal{E}}{\partial x}^{0} - \mu_p \mathcal{E} \frac{\partial P_n}{dx} + D_p \frac{\partial^2 P_n}{\partial x^2} + G_{Lp} - \frac{P_n - P_{no}}{\tau_p}$$

$$0 = +D_p \frac{\partial^2 P_n}{\partial x^2} + G_L - \frac{P_n - P_{no}}{\tau_p}$$

$$0 = D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n}{\tau_p} + G_L + \frac{P_{no}}{\tau_p}$$

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$$-\left(G_L + \frac{P_{n0}}{\tau_p}\right) = D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n}{\tau_p}$$

Charasthericstic equation of homogeneous:

$$D_p r^2 - \frac{1}{\tau_p} = 0$$

$$r = \pm \frac{1}{\sqrt{D_p \tau_p}}$$

$$r = \pm \frac{1}{L_p}$$

Thus:

$$P_n = A e^{x/L_p} + B e^{-x/L_p}$$

The particular solution is:

$$-\left(G_L + \frac{P_{n0}}{\tau_p}\right) = D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n}{\tau_p} \rightarrow P_n \text{ is a cte}$$

$$\downarrow$$

$$\frac{\partial^2 P_n}{\partial x^2} = 0$$

$$-\left(G_L + \frac{P_{n0}}{\tau_p}\right) = -\frac{P_n}{\tau_p}$$

$$G_L + \frac{P_{n0}}{\tau_p} = \frac{P_n}{\tau_p}$$

$$P_n = P_{n0} + \tau_p G_L$$

So; the complete solution:

$$P_n = A e^{x/L_p} + B e^{-x/L_p} + P_{n0} + \tau_p G_L$$

At $x \to \infty$

$$P_n(x \to \infty) = A \lim_{x \to \infty} e^{x/L_p} + P_{n0} + \tau_p G_L$$

$$P_{n0} + \tau_p G_L = P_{n0} + \tau_p G_L \qquad \text{only: } A = 0$$

Thus:

$$P_n = B e^{-x/L_p} + P_{n0} + \tau_p G_L$$

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When $x = 0$:

$$\left.\frac{\partial p_n}{\partial x}\right|_{x=0} = -\frac{B}{L_p} e^{-x/L_p}\Big|_{x=0}$$

$$D_p \left.\frac{\partial p_n}{\partial x}\right|_{x=0} = -\frac{B D_p}{L_p}$$

$$S_{xr}[p_n(0) - p_{no}] = -\frac{B D_p}{L_p}$$

$$B = -\frac{S_{xr} L_p}{D_p}[p_n(0) - p_{no}]$$

On the other hand, $x = 0, +\infty$ when:

$$p_n(0) = B e^{-x/L_p} + p_{no} + \tau_p G_L$$

$$p_n(0) = -\frac{S_{xr} L_p}{D_p}[p_n(0) - p_{no}] + p_{no} + \tau_p G_L$$

For find $p_n(0)$:

$$p_n(0) + \frac{S_{xr} L_p}{D_p} p_n(0) = \frac{S_{xr} L_p}{D_p} p_{no} + p_{no} + \tau_p G_L$$

$$D_p p_n(0) + S_{xr} L_p p_n(0) = S_{xr} L_p p_{no} + D_p p_{no} + D_p \tau_p G_L$$

$$p_n(0) = \frac{S_{xr} L_p p_{no} + D_p p_{no} + D_p \tau_p G_L}{D_p + S_{xr} L_p}$$

$$p_n(0) = \frac{(S_{xr} L_p + D_p)}{S_{xr} L_p + D_p} p_{no} + \frac{D_p \tau_p G_L}{D_p + S_{xr} L_p}$$

Thus:

$$B = -\frac{S_{xr} L_p}{D_p}\left[\frac{D_p \tau_p G_L}{D_p + S_{xr} L_p}\right] =$$

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Finally:

$$p_n(x) = B e^{-x/L_p} + p_{n0} + \tau_p G_L$$

$$p_n(x) = p_{n0} + \tau_p G_L - \frac{S_{lr} L_p \tau_p G_L}{D_p + S_{lr} L_p} e^{-x/L_p}$$

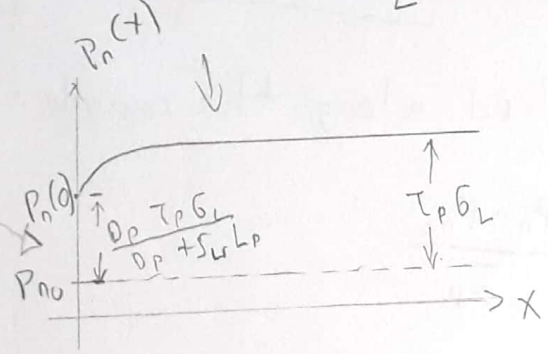
$$p_n(x) = p_{n0} + \tau_p G_L \left[1 - \frac{S_{lr} L_p}{D_p + S_{lr} L_p} e^{-x/L_p} \right]$$

$$\rightarrow L_p = \sqrt{D_p \tau_p}$$

$$D_p = \frac{L_p^2}{\tau_p}$$

$$p_n(x) = p_{n0} + \tau_p G_L \left[1 - \frac{S_{lr} L_p}{\frac{L_p^2}{\tau_p} + S_{lr} L_p} e^{-x/L_p} \right]$$

$$p_n(x) = p_{n0} + \tau_p G_L \left[1 - \frac{S_{lr} \tau_p}{L_p + S_{lr} \tau_p} e^{-x/L_p} \right]$$



At $x = 0$:

$$1 - \frac{S_{lr} L_p e^{-x/L_p}}{D_p + S_{lr} L_p} = \frac{D_p}{D_p + S_{lr} L_p}$$

When There is not recombination at the surface: $S_{lr} = 0$

$$p_n(x) = p_{n0} + \tau_p G_L = cte$$

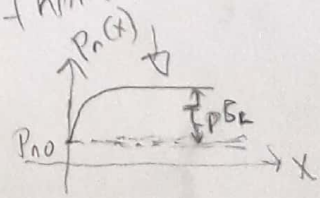
Equal to $p_n(x \rightarrow \infty) = p_{n0} + \tau_p G_L$

When $S_{lr} \rightarrow \infty$:

$$p_n(x) = p_{n0} + \tau_p G_L \left[1 - \frac{\tau_p e^{-x/L_p}}{\frac{L_p}{S_{lr}} \rightarrow 0 + \tau_p} \right]$$

$$p_n(x) = p_{n0} + \tau_p G_L \left(1 - e^{-x/L_p} \right)$$

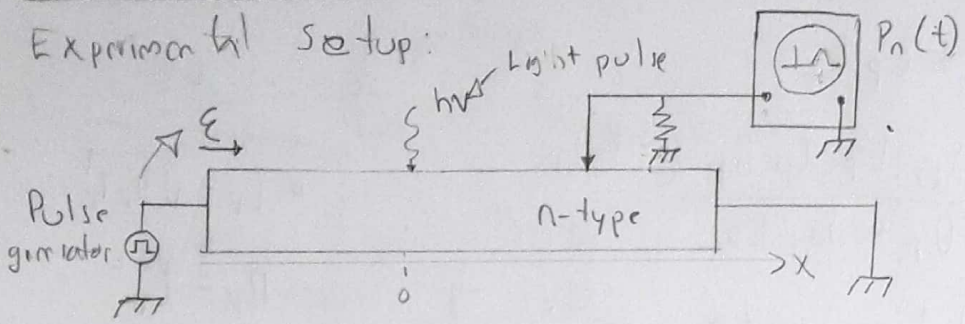
And at the surface: $x = 0$: $p_n(0) = p_{n0}$ it's thermal equilibrium



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2.4.3 The Haynes-Shockley Experiment

Experimental setup:



When localized light pulse at $x=0$, the transport equation after a pulse: $\rightarrow G_{L_P}=0$

$$\frac{\partial P_n}{\partial t} = -P_n \mu_p \frac{\partial \mathcal{E}}{\partial x}^{0} - \mu_p \mathcal{E} \frac{\partial P_n}{\partial x} + D_p \frac{\partial^2 P_n}{\partial x^2} + G_{L_P}^{0} - \frac{P_n - P_{n0}}{\tau_p}$$

$\rightarrow \rho_s = q(p - n + N_D^+ - N_A^-) = 0$

$$\frac{\partial \mathcal{E}}{\partial x} = \frac{\rho_s}{\varepsilon_s} = 0 \quad (s: \text{space})$$

$$\frac{P_n - P_{n0}}{\tau_p} = U = R_p - G_{th_p}$$

Thermal generation

If an electric field is applied along the sample: $\mathcal{E} \neq 0$

$$\frac{\partial P_n}{\partial t} = -\mu_p \mathcal{E} \frac{\partial P_n}{\partial x} + D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n - P_{n0}}{\tau_p}$$

A particular solution is:

$$P_n = P_{n0} \quad \therefore \quad \frac{\partial P_n}{\partial t} = \frac{\partial P_n}{\partial x} = \frac{\partial^2 P_n}{\partial x^2} = \frac{P_n - P_{n0}}{\tau_p} = 0$$

And the homogeneous solution must:

$$\frac{\partial P_n}{\partial t} + \mu_p \mathcal{E} \frac{\partial P_n}{\partial x} - D_p \frac{\partial^2 P_n}{\partial x^2} + \frac{P_n}{\tau_p} = 0$$

$$\frac{\partial P_n}{\partial t} = -\mu_p \mathcal{E} \frac{\partial P_n}{\partial x} + D_p \frac{\partial^2 P_n}{\partial x^2} - \frac{P_n}{\tau_p}$$

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The homogenous solution is:

$$p_n = \frac{N}{\sqrt{4\pi D_p t}} \exp\left[-\frac{(x-\mu_p \mathcal{E} t)^2}{4 D_p t} - \frac{t}{\tau_p}\right]$$

Where N: Number of electron-holes pairs generates per unit area in $x = 0$

Finally, the complet solution is (particular + homogenous)

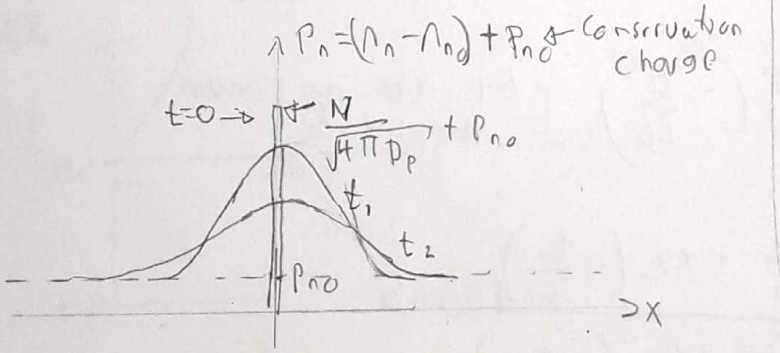
$$p_n = \frac{N}{\sqrt{4\pi D_p t}} \exp\left(-\frac{(x-\mu_p \mathcal{E} t)^2}{4 D_p t} - \frac{t}{\tau_p}\right) + p_{no}$$

If no field is applied along the sample, $\mathcal{E} = 0$:

$$p_n = \frac{N}{\sqrt{4\pi D_p t}} \exp\left(-\frac{x^2}{4 D_p t} - \frac{t}{\tau_p}\right) + p_{no}$$

$$p_n = \frac{N}{\sqrt{4\pi D_p t}} \exp\left(-\frac{x^2}{4 D_p t}\right) \exp\left(-\frac{t}{\tau_p}\right) + p_{no}$$

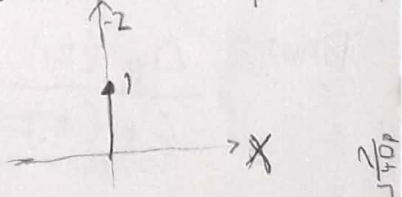
Gauss function around $x = 0$



At: $t \to 0$; $\sqrt{t} \to 0$

$$\frac{N}{\sqrt{t}} \exp\left(-\left(\frac{x}{\sqrt{4 D_p}\sqrt{t}}\right)^2\right)$$

Dirac delta function

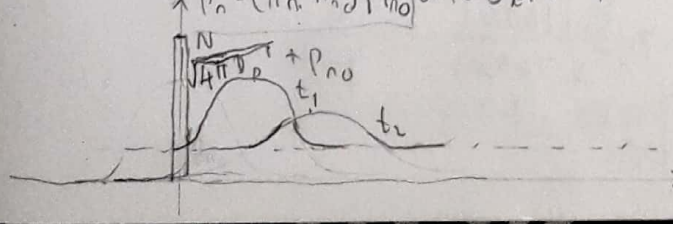


$$p_n = \frac{N}{\sqrt{4\pi D_p}} \frac{1}{\sqrt{t}} \left[\exp\left(-\left(\frac{x}{\sqrt{4 D_p}\sqrt{t}}\right)^2\right)\right]$$

If $\mathcal{E} > 0$:

$$p_n = \frac{N}{\sqrt{4\pi D_p t}} \exp\left(-\frac{(x-\mu_p \mathcal{E} t)^2}{4 D_p t}\right) \exp\left(\frac{-t}{\tau_p}\right) + p_{no}$$

Gauss function around $x = \mu_p \mathcal{E} t$



→ The excess minoritary carrier moves with drift velocity $\mu_p \mathcal{E}$

$$V_p = \mu_p \mathcal{E}$$

$$p_n = \frac{N}{\sqrt{4\pi D_p}} \left[\left(\frac{1}{\sqrt{t}}\right)^{4D_p} \exp\left(-\left(\frac{x}{\sqrt{4 D_p}\sqrt{t}}\right)^2\right)\right]$$

$$p_n(t \to 0, x = 0) = \frac{N}{\sqrt{4\pi D_p}} + p_{no}$$

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How: $v_p = \mu_p \mathcal{E}$ with a know sample length L, the drift field $\mathcal{E}$, and the time delay "t" between the applied electric pulse and the detected pulse:

$$\frac{L}{t} = \mu_p \mathcal{E}$$

$$\mu_p = \frac{L}{\mathcal{E} t}$$ } The Haynes-Shockley Experiment.

EXAMPLE 8

$\rightarrow \Delta p = p_n - p_{n_0}$ $\rightarrow t_1 = 100\mu s$ $\tau_p = ?$

$\rightarrow t_2 = 200\mu s$

$$\rightarrow \frac{\Delta p(t_1)}{\Delta p(t_2)} = 5$$

At the maximum amplitude:

$$\rightarrow \Delta p = p_n - p_{n_0} = \underbrace{\frac{N}{\sqrt{4\pi D_p t}}}_{\text{The gauss function is around } \mu_p \mathcal{E} t} \exp\left[- \frac{(x - \mu_p \mathcal{E} t)^2}{4 D_p t} - \frac{t}{\tau_p} \right]$$

$\rightarrow x = \mu_p \mathcal{E} t$

$$\Delta p = \frac{N}{\sqrt{4\pi D_p t}} \exp\left(-\frac{t}{\tau_p}\right)$$ } At the maximum amplitude.

Thus: $$\frac{\Delta p(t_1)}{\Delta p(t_2)} = \frac{\frac{1}{\sqrt{t_1}} \exp\left(-\frac{t_1}{\tau_p}\right)}{\frac{1}{\sqrt{t_2}} \exp\left(-\frac{t_2}{\tau_p}\right)} = 5$$

Therefore: $$5 = \sqrt{\frac{t_2}{t_1}} \exp\left(\frac{t_2 - t_1}{\tau_p}\right)$$

$$5\sqrt{\frac{t_1}{t_2}} = \exp\left(\frac{t_2 - t_1}{\tau_p}\right) = 5\sqrt{\frac{100\mu s}{200\mu s}} = \frac{5}{\sqrt{2}}$$

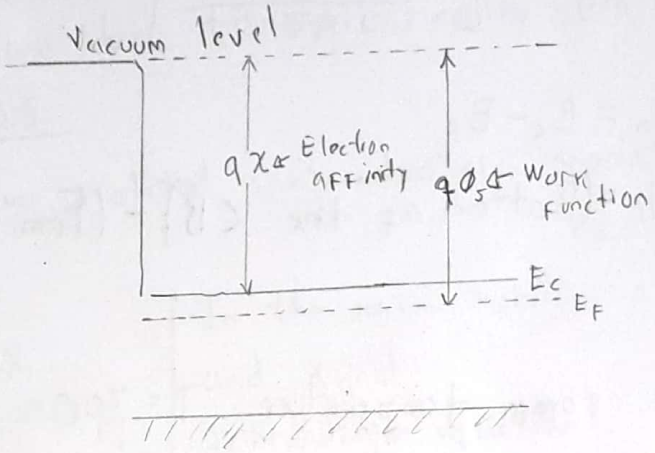
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$$\ln\left(\frac{5}{\sqrt{2}}\right) = \frac{t_2 - t_1}{\tau_p}$$

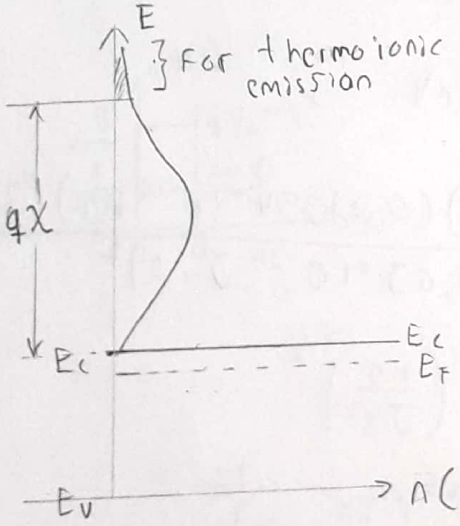
$$\tau_p = \frac{(200\mu s) - (100\mu s)}{\ln(5/\sqrt{2})} = 79\mu s$$

2.5 THERMOIONIC EMISSION PROCESS

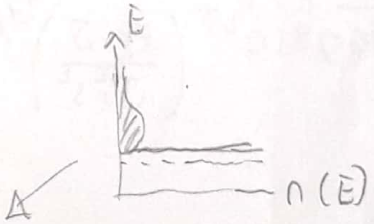
The band diagram of an isolated n-type semiconductor:



If the carrier concentration $n(E)$ in the conduction band:



$n(E)$ = Carrier concentration
= Electron distribution



Conduction Band

From the electron density above E_c:

$$n = \int_{E_c}^{\infty} N(E)F(E)\,dE = \int_{E_c}^{\infty} n(E)\,dE$$

$N(E)$: Density of states; $F(E)$: Fermi distribution

$$n = N_c \exp\left[\frac{-(E_c - E_F)}{kT}\right]$$

Effective density of states in the CB

$$N_c = 12\left(\frac{2\pi m_n kT}{h^2}\right)^{3/2}$$

For Si

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Thus, the electron density with energies above $q\chi$:

$$n_{th} = \int_{q\chi + E_c}^{\infty} n(E)\,dE = N_c \exp\left[-\frac{((E_c + q\chi) - E_F)}{kT}\right]$$

$$n_{th} = N_c \exp\left[-\frac{(q\chi + E_c - E_F)}{kT}\right]$$

$$n_{th} = N_c \exp\left[-\frac{q(\chi + V_n)}{KT}\right]$$

Where $qV_n = E_c - E_F$

$qV_n = (\text{Bottom of the CB}) - (\text{Fermi level})$

EXAMPLE 9

→ $n_{th} = ?$ → At room temperature: $T = 300K$ for n-type Si

$q\chi = 4.05\,eV$ → $qV_n = 0,2\,eV$

$n_{th} = ?$ → $T = 300K$ n-type Si

$q\chi = 10.6\,eV$ → $qV_n = 0,2\,eV$

$$\rightarrow N_c = 12\left(\frac{2\pi m_n kT}{h^2}\right)^{3/2} = 12\left[\frac{2\pi(0,26)(0.91094\times10^{-30}\,kg)(25.8\,meV)}{(6.63\times10^{-34}\,J.s)^2}\right]^{3/2}$$

$$N_c = 1.89\times10^{25}\left(\frac{kg\,J}{J^2 s^2}\right)^{3/2} = 1.89\times10^{25}\left(\frac{kg}{Js^2}\right)^{3/2}$$

$$= 1.89\times10^{25}\left(\frac{kg}{kg\frac{m^2}{s^2}s^2}\right)^{3/2}$$

$$= \frac{1.89\times10^{25}}{m^3} = \frac{1.89\times10^{25}}{(100\,cm)^3}$$

$$= 1.89\times10^{19}\,cm^{-3}$$

From Sze: $N_c = 2.86\times10^{19}\,cm^{-3}$

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$$\rightarrow n_{th}(q\chi = 4.05eV) = 2.86\times10^{19}cm^{-3}\exp\left[-\frac{(4.05eV)+0.2eV}{0.0259eV}\right]$$

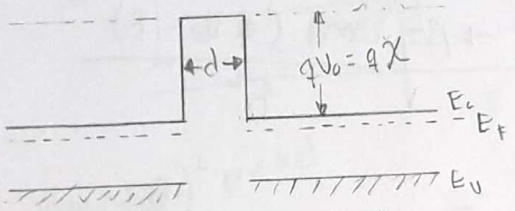
$n_{th} = 10^{-52}cm^{-3} \approx 0$ ← There is no emission electrons into vacuum

$$\rightarrow n_{th}(q\chi = 0.6eV) = 2.86\times10^{19}cm^{-3}\exp\left[-\frac{(0.6eV)+(0.2eV)}{0.0259eV}\right]$$

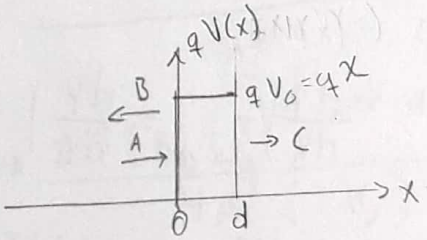
$n_{th} = 10^{6}cm^{-3}$ ← Substantial number of electrons can be thermoionically emitted

2.6 TUNNELING PROCESS

The band diagram of two isolated semiconductors separated by a distance d: (n-type)



One dimensional potential barrier:



In the region where $q(V_x) = 0$, $x \leq 0$ and $x \geq d$:

Schrödinger equation: ← kinetic energy

$$-\frac{\hbar^2}{2m_n}\frac{d^2\psi}{dx^2} + q\cancel{V_0}^{0}\psi = E\psi$$

↑ Effective mass

← Characteristic eq.

$$\rightarrow -\frac{\hbar^2}{2m}r^2 = E$$

$$r = \pm\sqrt{\frac{2m_nE}{\hbar^2}}\,j \qquad \rightarrow \kappa = \sqrt{\frac{2m_nE}{\hbar^2}}$$

$$r = \pm\kappa j$$

Thus:

→ For $x \leq 0$: $\psi = \underbrace{Ae^{j\kappa x}}_{\text{Incident particle wave Function}} + \underbrace{Be^{-j\kappa x}}_{\text{Reflect particle wave Function}}$

→ For $x \geq d$: $\psi = \underbrace{Ce^{j\kappa x}}_{\text{Transmitted wave Function}} + \cancel{De^{-j\kappa x}}^{0} = Ce^{j\kappa x}$

Escaneado con CamScanner

Inside the potential barrier, $0 \leq x \leq d$

$$-\frac{\hbar^2}{2m_n}\frac{d^2\psi}{dx^2} + qV_0\psi = E\psi$$

$$\frac{d^2\psi}{dx^2} = \frac{2m_n(qV_0 - E)}{\hbar^2}\psi$$

Characteristic equation:

$$r = \pm\sqrt{\frac{2m_n(qV_0 - E)}{\hbar^2}} =$$

The solution for $E < qV_0$, $\Leftarrow N_0$ thermionic process

$$r = \pm\beta \quad \rightarrow \beta = \sqrt{\frac{2m_n(qV_0 - E)}{\hbar^2}}$$

So:

$$\rightarrow \boxed{\psi(x) = Fe^{\beta x} + Ge^{-\beta x} \quad 0 \leq x \leq d}$$

The continuity at $x = 0$:

$$Ae^{jkx} + De^{-jkx} = Fe^{\beta x} + Ge^{-\beta x}$$

$$A + B = F + G$$

$$\left.\frac{d\psi}{dx}\right|_{x\to 0^-} = \left.\frac{d\psi}{dx}\right|_{x\to 0^+}$$

$$Ajke^{jkx} + Bjke^{-jkx} = F\beta e^{\beta x} - G\beta e^{-\beta x}$$

$$Ajk - Bjk = F\beta - G\beta$$

$$\rightarrow \begin{aligned} Ajk + Bjk &= Fjk + Gjk \\ Ajk - Bjk &= F\beta - G\beta \end{aligned}$$

$$2Ajk = F(jk + \beta) + G(jk - \beta)$$

$$A = \frac{F(jk + \beta) + G(jk - \beta)}{2jk}$$

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The continuity at $x = d$:

$$F e^{\beta d} + G e^{-\beta d} = C e^{jkd} \qquad F\beta e^{\beta d} - G\beta e^{-\beta d} = Cjk e^{jkd}$$

$$\rightarrow \begin{array}{l} F\beta e^{\beta d} + G\beta e^{-\beta d} = C\beta e^{jkd} \\ F\beta e^{\beta d} - G\beta e^{-\beta d} = Cjk e^{jkd} \end{array}$$

$$2F\beta e^{\beta d} = C((kj+\beta) e^{jkd}$$

$$F = \frac{C((kj+\beta) e^{jkd}}{2\beta e^{\beta d}}$$

$$\rightarrow \begin{array}{l} -F\beta e^{\beta d} - G\beta e^{-\beta d} = -C\beta e^{jkd} \\ F\beta e^{\beta d} - G\beta e^{-\beta d} = Cjk e^{jkd} \end{array}$$

$$-2G\beta e^{-\beta d} = C((kj-\beta) e^{jkd}$$

$$G = \frac{C((kj-\beta) e^{jkd}}{-2\beta e^{-\beta d}}$$

Thus:

$$A = \left[\frac{C((kj+\beta) e^{jkd}}{2\beta e^{\beta d}}\right](kj+\beta) + \left[\frac{C((kj-\beta) e^{jkd}}{-2\beta e^{-\beta d}}\right](kj-\beta)$$

$$\frac{A}{C} = \frac{\frac{(kj+\beta)^2 e^{jkd}}{2\beta e^{\beta d}} + \frac{(kj-\beta)^2 e^{jkd}}{2\beta e^{-\beta d}}}{2kj}$$

$$\frac{A}{C} = \frac{2\beta(kj+\beta)^2 e^{jkd-\beta d} - 2\beta(kj-\beta)^2 e^{jkd+\beta d}}{4\beta^2(2kj)}$$

$$\frac{A}{C} = \frac{2\beta e^{jkd}\left[(kj+\beta)^2 e^{-\beta d} - (kj-\beta)^2 e^{\beta d}\right]}{8\beta^2 kj}$$

$$\frac{A}{C} = \frac{e^{jkd}\left[(-k^2+2kj\beta+\beta^2) e^{-\beta d} - (-k^2-2kj\beta+\beta^2) e^{\beta d}\right]}{4\beta kj}$$

$$\frac{A}{C} = e^{jkd}\left[\frac{(k^2-\beta^2)(2\sinh\beta d)}{4\beta kj} + \frac{2kj\beta(2\cosh\beta d)}{4\beta kj}\right]$$

$$\frac{A}{C} = e^{jkd}\left[\frac{(\beta^2-k^2)\sinh\beta d}{2\beta k} j + \cosh\beta d\right]$$

Escaneado con CamScanner

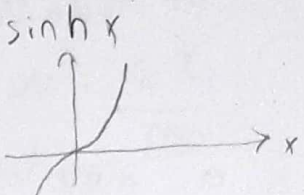
$$\left|\frac{A}{C}\right|^2 = \frac{(\beta^2 - k^2)^2}{4\beta^2 k^2} \sinh^2 \beta d + \cosh^2 \beta d$$

Escaneado con CamScanner

Transmission coefficient:

$$T = \frac{|Ce^{ikx}|^2}{|Ae^{ikx}|^2} = \frac{|A|^2 |e^{ikx}|^2}{|C|^2 |e^{ikx}|^2} \to 1$$

$$T = \left(\frac{C}{A}\right)^2 = \left[1 + \frac{(qV_0 \sinh \beta d)^2}{4E(qV_0 - E)}\right]^{-1}$$



$$T = \left(\frac{C}{A}\right)^2 = \left[\frac{4E(qV_0 - E)}{4E(qV_0 - E) + (qV_0 \sinh \beta d)^2}\right]$$

* I_F: $\beta d = 1 = \sqrt{\frac{2m_n(qV_0 - E)}{\hbar^2}}\, d$

$\to \sinh \beta d = \frac{e^1 - e^{-1}}{2} = 1{,}543 \approx \frac{e^1}{2} = 1{,}359$

* I_F: $\beta d > 1$:

$\to \sinh \beta d \approx \frac{e^{\beta d}}{2}$

So:

$$T \approx \frac{4E(qV_0 - E)}{4E(qV_0 - E) + \left[qV_0\left(\frac{e^{\beta d}}{2}\right)\right]^2}$$

$$T \propto \frac{1}{e^{2\beta d}}$$

$$T \propto e^{-2\beta d}$$

$$T \propto \exp\left[-2d\sqrt{\frac{2m_n(qV_0 - E)}{\hbar^2}}\right]$$

For finite transmission coefficient: small tunneling distance d
low potential barrier qV_0
small effective mass M_n

Escaneado con CamScanner

2.7 SPACE-CHARGE EFFECT

The space charge density:

$$\rho = q(p - n + N_D^+ - N_A^-)$$

$p - n$: carrier concentration; $N_D^+ - N_A^-$: Impurity concentrations

In a neutral region of a semiconductor:

$$\rightarrow p = N_A^- \qquad \rightarrow n = N_D^+$$

$$\rightarrow \rho = 0$$

In a n-type semiconductor:

$$\rightarrow N_D^+ >> N_A^- = p \approx 0$$

$$\rightarrow n = N_D^+$$

$$\rightarrow \rho = 0$$

If we inject electrons:

$$\rightarrow n >> N_D^+$$

$$\rightarrow \rho = q(\cancel{p}^{\,0} - n + \cancel{N_D^+}^{\,0} - \cancel{N_A^-}^{\,0})$$

$$\rho \approx -qn$$

Space charge effect:

Poisson's equation:

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

$$\Downarrow$$

$$\mathcal{E} = \mathcal{E}(x)$$

* Space-charge limited current: The current is dominated for:

$J = J_{drift}$ of n (Injected carriers)

$J = -qnv$

1) In the constant-mobility regime: $v \propto \mathcal{E}$

$\rightarrow v = -\mu \mathcal{E}$

↳ For electrons

$\rightarrow J = -q n v$

Mobility:

$$\mu = \frac{q \tau_c}{m_n}$$

Ohm's law:

$$J = q n \mu \mathcal{E}$$

Poisson's equation:

$$\rightarrow \frac{d\mathcal{E}}{dx} = \frac{|\rho|}{\varepsilon_s}$$ ←?

$$\frac{d\mathcal{E}}{dx} = -\frac{qn}{\varepsilon_s}$$

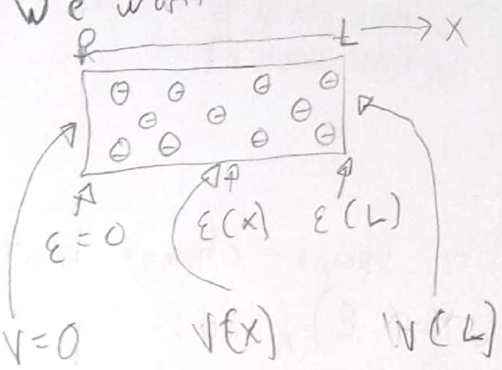
$$\frac{d\mathcal{E}}{dx} = \left(\frac{J}{\mu \mathcal{E}}\right)\frac{1}{\varepsilon_s}$$

$$\int_0^{\mathcal{E}} d\mathcal{E} = \int_0^x \frac{J}{\mu \varepsilon_s} dx$$

$$\frac{\mathcal{E}^2}{2} = \frac{J}{\mu \varepsilon_s} x$$

$$\mathcal{E} = \pm\sqrt{\frac{2Jx}{\mu \varepsilon_s}}$$

We want: $\xrightarrow{J}$



The current go from left to right: $\xrightarrow{\mathcal{E}}$

$$\mathcal{E} = \sqrt{\frac{2Jx}{\mu \varepsilon_s}} = \sqrt{\frac{2J}{\mu \varepsilon_s}}\, x^{1/2}$$

From: $\mathcal{E} = -\nabla V$

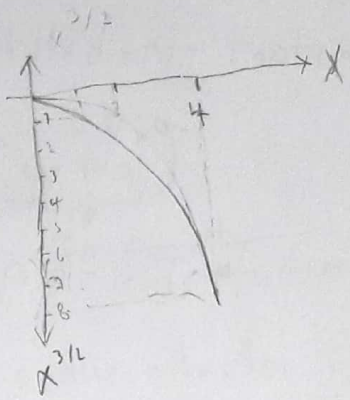
$$\mathcal{E} = -\frac{dV}{dx}$$

$$V = cte\,(L - x$$

$$\int_0^V dV = \int_0^x \mathcal{E}\, dx = \int_0^x \sqrt{\frac{2J}{\mu \varepsilon_s}}\, x^{1/2} dx = \sqrt{\frac{2J}{\mu \varepsilon_s}}\, \frac{2}{3} x^{3/2}$$

Escaneado con CamScanner

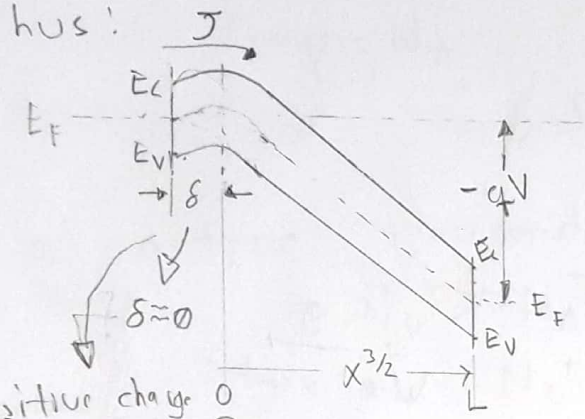
$$V = -\frac{2}{3}\sqrt{\frac{2J}{\varepsilon_s M}}\, x^{3/2}$$



For $x = L$

$$V(L) = -\frac{2}{3}\sqrt{\frac{2J}{\varepsilon_s M}}\, L^{3/2}$$

Thus:



Positive charge 0 in the surface?

$$\rightarrow V = -\frac{2}{3}\sqrt{\frac{2J}{\varepsilon_s M}}\, L^{3/2}$$

$$V^2 = \frac{4}{9}\frac{(2J)}{\varepsilon_s M} L^3$$

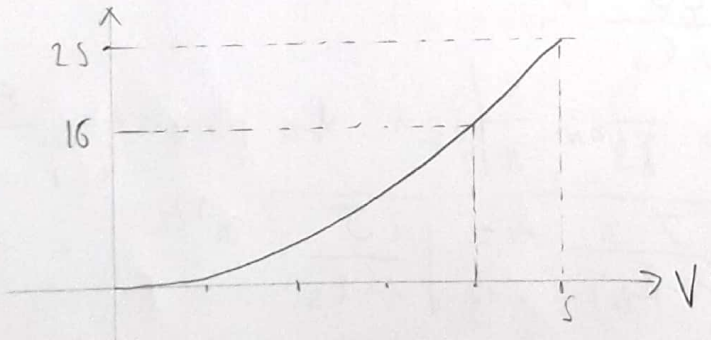
$$J = \frac{9\varepsilon_s M V^2}{8 L^3}$$

$$\rightarrow J \sim V^2 \quad \rightarrow J \sim \frac{1}{L^3}$$

The space-charge-limited current in the constant-mobility ($V \propto \mathcal{E}$), so:

velocity

$$J \Big/ \left(\frac{9\varepsilon_s M}{8 L^3}\right)$$



Escaneado con CamScanner

2) In the velocity-saturation regime:

→ v_s = constant

$v_s \neq F(\mathcal{E})$

$v_s \neq \mu \mathcal{E}$

From: $J = -q n v_s$

↑ Injected carriers

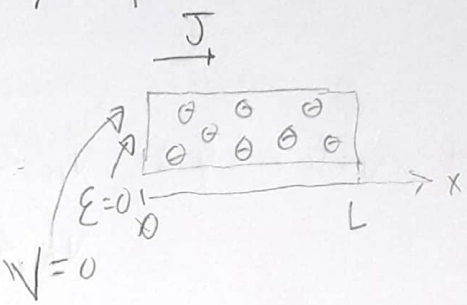
$$q n = \frac{-J}{v_s}$$

Poisson's equation:

$$\frac{d\mathcal{E}}{dx} = \frac{|\rho|}{\varepsilon_s} \quad ?$$

→ $\rho = -qn$

$$\frac{d\mathcal{E}}{dx} = \frac{|-qn|}{\varepsilon_s}$$



$$\frac{d\mathcal{E}}{dx} = \frac{J}{v_s \varepsilon_s}$$

↑ constant ∴ $\neq f(\mathcal{E})$

$$\int_0^{\mathcal{E}} d\mathcal{E} = \int_0^x \frac{J}{v_s \varepsilon_s} dx$$

$$\mathcal{E} = \frac{J x}{v_s \varepsilon_s}$$

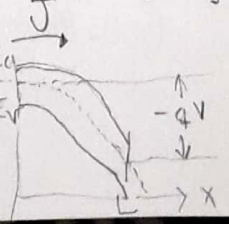
→ $\mathcal{E} = -\nabla V$

$$\mathcal{E} = -\frac{dV}{dx}$$

$$\int_0^V dV = \int_0^x \mathcal{E}\, dx$$

$$V = \int_0^x \frac{-J x}{v_s \varepsilon_s} dx = \frac{-J x^2}{2 v_s \varepsilon_s}$$

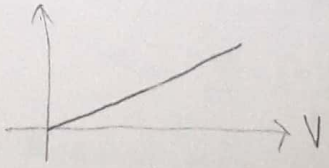
$$V(L) = -\frac{J L^2}{2 \varepsilon_s v_s}$$



Space-charge-limited current in the velocity-saturation current:

$$|J| = \frac{2 \varepsilon_s v_s V}{L^2}$$

→ $|J| \sim V$ → $|J| \sim \frac{1}{L^2}$



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2.8 HIGH-FIELD EFFECTS

*At low electric-fields:

Drift velocity:

$\rightarrow v_n = -\mu_n \mathcal{E}$ ← Ohm's law

$v_n = -\left(\frac{q\tau_c}{m_n}\right)\mathcal{E}$

$\rightarrow \mu_n = \frac{q\tau_c}{m_n}$

$\tau_c = \frac{\mu_n m_n}{q}$

$\tau_c \neq f(\mathcal{E})$

Between collision: Mean Free time

Drift velocity: $v_n \ll v_{th}$

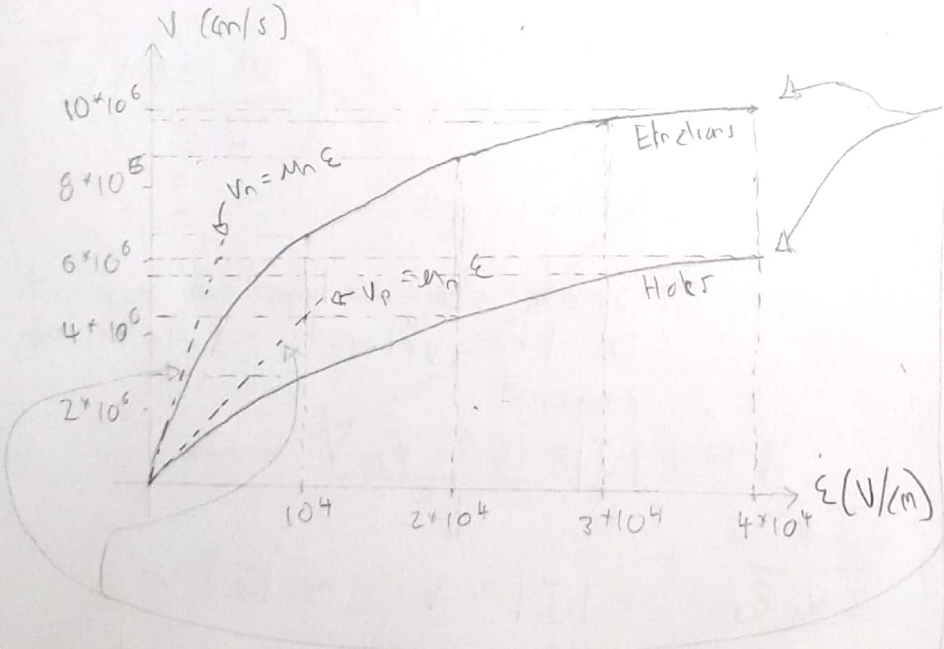
$v_n \ll 10^7 cm/s$

$\frac{1}{2} m_n v_{th}^2 = \frac{3}{2} kT$

$v_{th} = \sqrt{\frac{3kT}{m_n}}$

$v_{th} \approx 10^7 cm/s$

* Measured drift velocities in Si: (300K)



Real

From linear model (ideal):

$\rightarrow \mu_n \approx \frac{v_n}{\mathcal{E}} \approx \frac{6*10^6 cm/s}{0.5*10^4 V/cm}$

$\mu_n = 1200 \frac{cm^2}{Vs}$

From graphs: $N_D \approx 10^{16} cm^{-3}$

$\rightarrow \mu_p = \frac{v_p}{\mathcal{E}} \simeq \frac{4*10^6 cm/s}{0.9*10^4 V/cm} = 444 \frac{cm^2}{Vs}$

From graphs: $N_A \simeq 10^{16} cm^{-3}$

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Empirical expression:

$$V_{n/p} = \frac{V_s}{\left[1 + \left(\frac{\mathcal{E}_{0n/p}}{\mathcal{E}}\right)^{\gamma_{n/p}}\right]^{1/\gamma_{n/p}}}$$

→ V_s = Saturation velocity $\approx V_{th}$

$V_s = 10^7$ cm/s

→ Constant $\mathcal{E}_0$:

→ $\mathcal{E}_{0n} = 7 * 10^3$ V/cm → $\mathcal{E}_{0p} = 2 * 10^4$ V/cm

in high-purity silicon materials

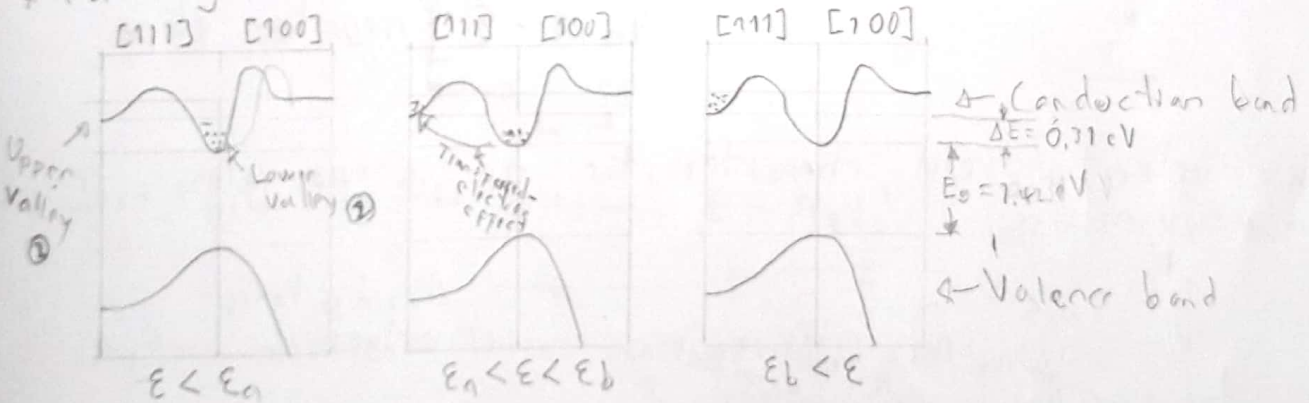
→ $\gamma_n = 2$ → $\gamma_p = 1$

At sufficiently large fields:

$$V_{n/p} = \lim_{\mathcal{E} \to \infty} \frac{V_s}{\left[1 + \left(\frac{\mathcal{E}_{0n/p}}{\mathcal{E}}\right)^{\gamma_{n/p}}\right]^{1/\gamma_{n/p}}}$$

$V_{n/p} \to V_s$

* For gallium arsenide: Electron distributions.



The steady-state conductivity of the n-type GaAs:

→ $\sigma = q(\mu_1 n_1 + \mu_2 n_2)$

→ $\sigma = q n \bar{\mu}$

→ $n = n_1 + n_2$

→ $\bar{\mu} = \frac{\mu_1 n_1 + \mu_2 n_2}{n}$

Average mobility

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Drift velocity:

$v_n = \bar{\mu} \mathcal{E}$

$\mu_n = \frac{q \tau_c}{m_n}$

For $0 < \mathcal{E} < \mathcal{E}_a$:

$\rightarrow n_2 \approx 0$,

$n \simeq n_1$

$\rightarrow v_n \approx \mu_1 \mathcal{E}$

For $\mathcal{E}_a < \mathcal{E} < \mathcal{E}_b$:

$\rightarrow n = n_1 + n_2$

$\rightarrow v_n = \frac{\mu_1 n_1 + \mu_2 n_2}{n_1 + n_2} \mathcal{E}$

For $\mathcal{E}_b < \mathcal{E}$:

$\rightarrow n_1 \approx 0$

$n = n_2$

$\rightarrow v_n \approx \mu_2 \mathcal{E}$

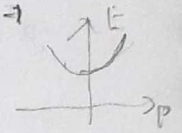
Like: $\text{concavity}_1 > \text{concavity}_2 \rightarrow m_1 < m_2 \rightarrow \mu_1 > \mu_2$

$\Downarrow$

$\mu_1 \mathcal{E} > \mu_2 \mathcal{E}$

For electron:

$m_n = \left(\frac{d^2 E}{dp^2} \right)^{-1}$



$m_n \propto \frac{1}{\text{concavity}}$

So while $\mathcal{E}_a < \mathcal{E} < \mathcal{E}_b$: IF $\mathcal{E}$ increasing, it could:

$\mu_1 n_1$ $\rightarrow$ $\mu_2 n_2$ and $\mu_2 < \mu_1$

$\Downarrow$

$n_2 > n_1$

$\Downarrow$

$\mu_1 n_1 + \mu_2 n_2$ decreasing

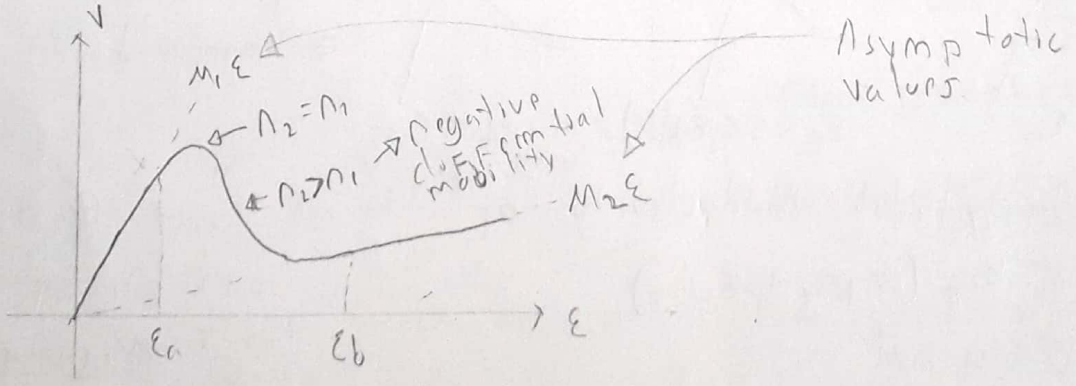
$\Downarrow$

$\bar{\mu}$ decreasing $\left(\frac{\partial \bar{\mu}}{\partial \mathcal{E}} < 0 \right)$

$\Downarrow$

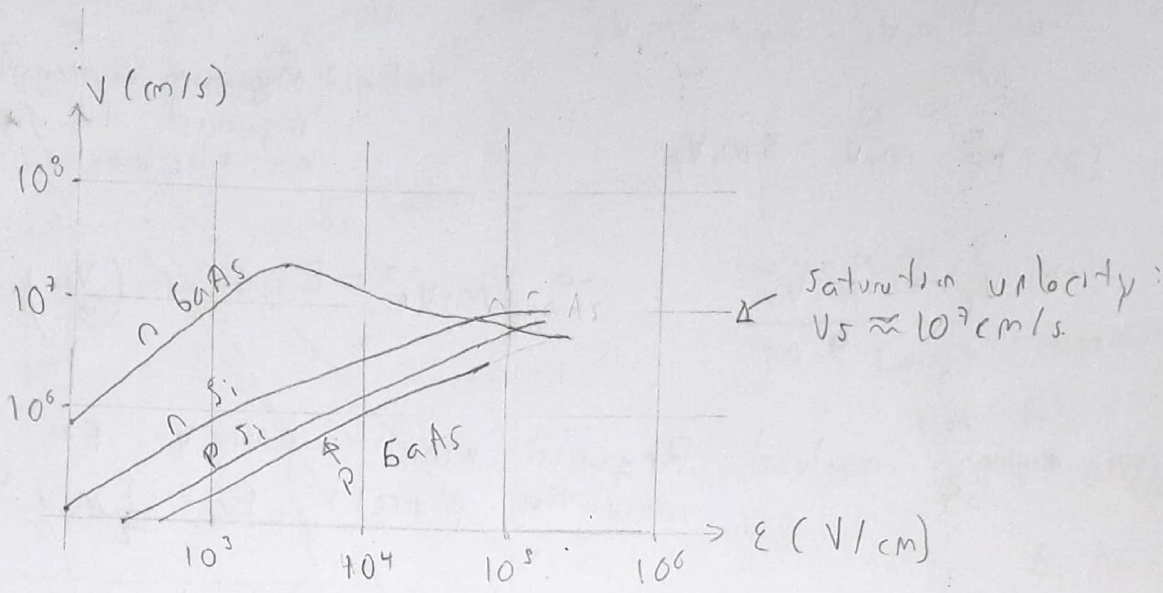
$v_n \propto -\mathcal{E}$ } negative slope

One possible velocity-field characteristic of a two-valley semiconductor:



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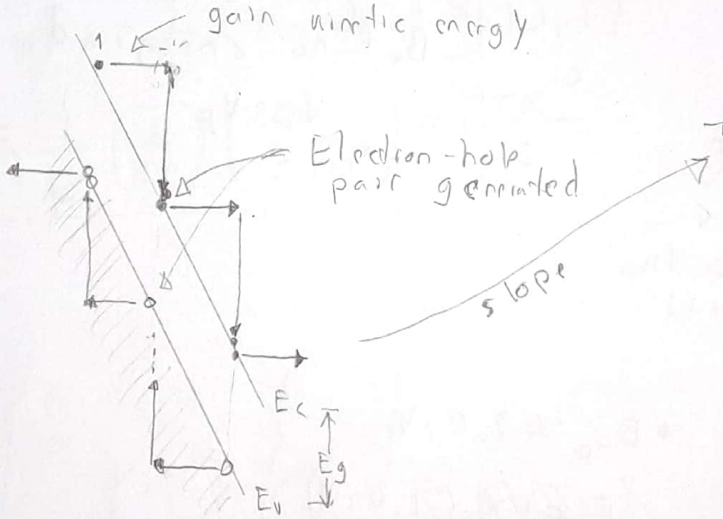
Thus the measured data:



* Avalanche process: High electric field: $V = V_s \neq f(\varepsilon)$

$\varepsilon \rightarrow V_s$

Energy band diagram: For the avalanche process → Impact ionization process



→ $\frac{dV}{dx} < 0$

→ $\varepsilon = -\frac{dV}{dx} = -\nabla V$

slope

$\varepsilon > 0$

ε

Just prior to the collision: $E = \frac{m_1 V_s^2}{2}$ $\quad p = m_1 V_s$

After collision: Three carriers: ① Original electro
② and ③: Electron-hole pair.

If ①, ②, ③ have: same effective mass: m_1
same kintic energy: $\frac{1}{2} m_1 V_F^2$
same momentum: $m_1 V_F$

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To conserve energy and momentum:

$\rightarrow \frac{1}{2} m_1 V_s^2 = E_g + \frac{3}{2} m_1 V_F^2$

$\rightarrow m_1 V_s = 3 m_1 V_F$

$\rightarrow E_g$ = Minimum energy required to generated an electron-hole pair.

So:

$\rightarrow V_F^2 = \frac{m_1^2 V_s^2}{9 m_1^2}$

$\rightarrow \frac{1}{2} m_1 V_s^2 = E_g + \frac{3}{2} m_1 \left(\frac{V_s^2}{9}\right)$

Ionization energy involved: Required kinetic energy for the ionization process: $E_0 = \frac{1}{2} m_1 V_s^2$

$E_0 = E_g + \frac{1}{2} m_1 V_s^2 / 3$

$E_0 = E_g + \frac{E_0}{3}$

$\frac{2E_0}{3} = E_g$

$E_0 = \frac{3}{2} E_g = 1.5 E_g$

Without avanlanch process:

$\frac{1}{2} m_1 V_s^2 = \frac{1}{2} m_1 V_F^2$

$V_s^2 = V_F^2$

$V_s = \pm V_F$

But no change the direction:

$V_s = V_F$

Thus: $E_0 > E_g \rightarrow$ ionization process

For silicon:

$\rightarrow E_{0n} \approx 1.6 eV \quad \rightarrow E_{0p} \approx 2.0 eV$

$E_{0n} \approx X \frac{(1.6 eV)}{(1.12 eV)} \qquad E_{0p} \approx X \frac{(2.0 eV)}{(1.12 eV)}$

$E_{0n} \approx 1.42 E_g \qquad E_{0p} \approx 1.79 E_g$

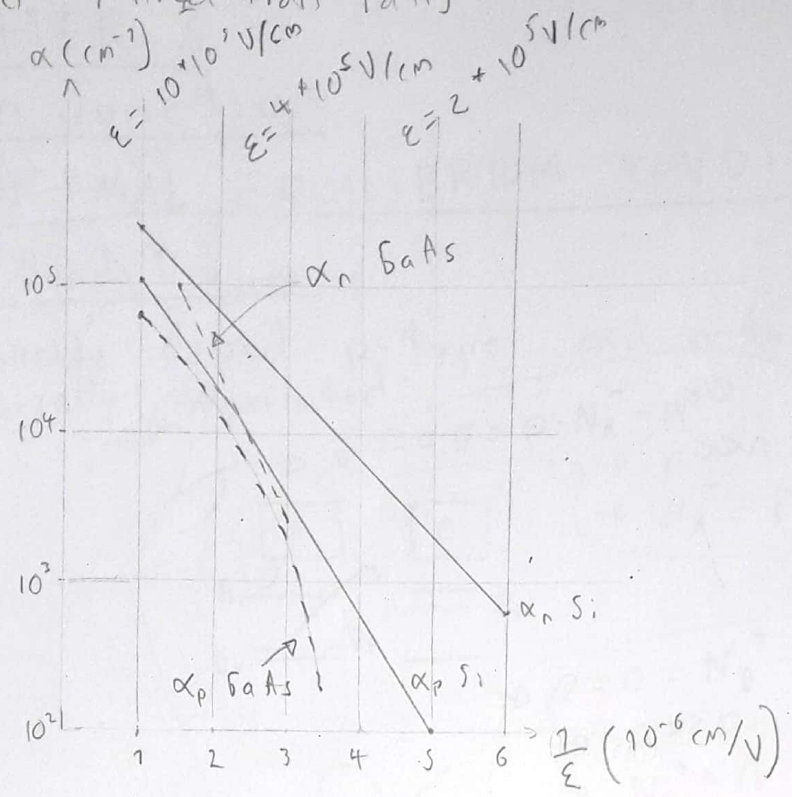
Ionization rate: $\alpha_n = \frac{\#\ \text{electron-hole pair generated}}{\text{Distance traveled by an electron}}$

$\alpha_p = \frac{\#\ \text{electron-hole pair generated}}{\text{Distance traveled by an hole}}$

$[\alpha] = cm^{-1}$

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Measured ionization rates



$\alpha \propto \frac{1}{(1/\varepsilon)}$

$\alpha \propto \varepsilon$

Large ionization rate:

For Si:

$\alpha_n(3 * 10^5 V/cm) \approx 10^4 cm^{-1}$

For GaAs:

$\alpha_n(4 * 10^5 V/cm) = 10^4 cm^{-1}$

Electron – hole pair generation rate avalanche process:

$$G_A = \frac{1}{q}(\alpha_n |J_n| + \alpha_p |J_p|)$$

$$\rightarrow [G_A] = \frac{1}{C}\left(\frac{1}{cm}\right)\left(\frac{C}{s\,cm^2}\right) = \frac{cm^{-3}}{s}$$

Problems

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